

Selecting Directors Using Machine Learning*

Isil Erel

Fisher College of Business, Ohio State University, NBER, and ECGI

Léa H. Stern

Foster School of Business, University of Washington

Chenhao Tan

University of Chicago

Michael S. Weisbach

Fisher College of Business, Ohio State University, NBER, and ECGI

Can algorithms assist firms in their decisions on nominating corporate directors? Directors predicted by algorithms to perform poorly indeed do perform poorly compared to a realistic pool of candidates in out-of-sample tests. Predictably bad directors are more likely to be male, accumulate more directorships, and have larger networks than the directors the algorithm would recommend in their place. Companies with weaker governance structures are more likely to nominate them. Our results suggest that machine learning holds promise for understanding the process by which governance structures are chosen and has potential to help real-world firms improve their governance. (*JEL* C10, C45, G30, M12, M14, M51)

Received May 12, 2019; editorial decision March 7, 2021 by Editor Itay Goldstein. The authors have furnished an Internet Appendix, which is available on the Oxford University Press Web site next to the link to the final published paper online.

A company's board of directors is legally responsible for managing the company. In principle, the board of directors reports to the shareholders and

* We thank two anonymous referees; Renée Adams and Reena Aggarwal (who graciously shared data), Lucian Bebchuk, Philip Bond, Lisa Cook, Ran Duchin, Daniel Ferreira (discussant), Fabrizio Ferri, Itay Goldstein (editor), Shan Ge, Jarrad Harford, Ben Hermalin, Joan MacLeod Heminway, Joshua Lee (discussant), Nadya Malenko (discussant), Jordan Nickerson (discussant), Miriam Schwartz-Ziv, Anil Shivdasani, Tracy Yue Wang (discussant), Ayako Yasuda, and Luigi Zingales (discussant); and conference and seminar participants at London School of Economics, Michigan State-UIUC Virtual Finance Seminar, University of North Carolina, Nanyang Tech, National University of Singapore, Northeastern, Ohio State, SMU, Tennessee, Utah, Washington, the 2017 Pacific Northwest Finance Conference, the 2017 WAPFIN Conference at NYU Stern, the 2017 NABE TEC Conference, the 2018 University of Miami-AFFECT Conference, the 2018 Drexel Corporate Governance Conference, the 2018 ICWSM BOD Workshop, the 2018 NBER Economics of AI Conference, the 2018 Washington University Olin Corporate Finance Conference, the 2019 AFA Annual Meeting, the 2019 NBER Big Data Conference, the 2019 Conference on Emerging Technologies in Accounting and Financial Economics at USC, the 2019 Wine Country Finance Conference, and the Machine Learning Day Conference at Georgetown. Special thanks to Ronan Le Bras of the Paul Allen Institute for AI for providing invaluable help throughout the project. Send correspondence to Léa Stern, leastern@uw.edu.

represents their interests. In practice, however, there is much variation in director quality and the extent to which they serve shareholders' interests.¹ Many of the concerns about boards come from the director selection process, which is one of the most important yet least studied questions in corporate governance.² Aside from occasional proxy contests, shareholders have virtually no control over the choice of the directors whose mandate is to represent their interests.

In this paper, we consider a potential alternative approach to selecting directors: one that uses algorithms that rely on data on firms, current board members, and the attributes of potential directors, to identify the quality of directors being considered for a given firm's board. Because boards must make predictions about the way that potential nominees will perform, the selection of directors is essentially a prediction problem. And while "traditional" econometrics is typically designed for estimating structural parameters and drawing causal inferences, machine learning algorithms are substantially better at making predictions. These algorithms are designed to maximize out-of-sample predictive accuracy by avoiding overfitting and by not being constrained by specific parametric assumptions or restrained in the number of covariates. They are particularly useful in applications when there is no clear guidance for how to build expectations. Because there is no one-size-fits-all "good" governance, many covariates potentially matter and may interact in nonlinear ways when predicting director performance. For this reason, we wish to rely on a rigorous data-driven model selection that produces more accurate predictions.³

We construct a large database of publicly traded U.S. firms and independent directors appointed between 2000 and 2014. We employ several machine learning algorithms designed to predict director performance using director-, board-, and firm-level data available to the nominating committee at the time of the nominating decision. We compare the algorithms' selections of directors to the ones actually chosen by firms. The discrepancies between firms' actual choices of directors and the choices based on the predictions from our algorithms allow us to characterize which individual features are overrated by decision makers. As such, the algorithms' predictions can provide insights into the decision-making process that governs the selection of corporate directors.

A crucial element of any algorithm designed to select valuable independent directors is a process for assessing a director's performance in a particular firm. The task of measuring the performance of an individual director is challenging. Directors generally act collectively on the board and it is usually impossible for

¹ See Hermalin and Weisbach (2003), Adams, Hermalin, and Weisbach (2010), and Adams (2017) for surveys.

² Berle and Means (1932) originally raised concerns about the director selection process. Hermalin and Weisbach (1998) present a formal model in which directors are not selected optimally from shareholders' perspectives. Shivdasani and Yermack (1999), Kramarz and Thesmar (2013), and Cai, Nguyen, and Walking (2017) provide evidence supporting this view.

³ See Athey and Imbens (2017) and Mullainathan and Spiess (2017).

a researcher to ascertain the actions of any director. Hart and Zingales (2017) emphasize that directors' fiduciary duty is to represent the interests of the firm's shareholders. Their popularity among shareholders is thus a natural metric for evaluating them. For that reason, our main measure of director performance is based on levels of shareholder support in annual director reelections. Voting based measures are individualized, market-based measures of performance that capture investors' preferences. Therefore, we task the algorithm with predicting the average excess voting support relative to the slate of directors up for reelection over the first 3 years of director tenure.

We employ several machine learning algorithms to predict the performance of a potential director at a particular company, taking into consideration who is currently sitting on the board. Using our sample of public firms, we train each algorithm on a training set (directors appointed between 2000 and 2011), and then compare the predictions to the observed out-of-sample data using a test set (directors appointed between 2012 and 2014).

We find that these algorithms make accurate out-of-sample predictions of the distribution of outcomes. The directors the algorithms predicted would poorly did worse on average than the directors the algorithms predicted would do well. In comparison, the directors predicted to do poorly by an Akaike information criterion (AIC)-selected ordinary least squares (OLS) model do not have worse performance out of sample than those it predicted would do well.

We ensure that the out-of-sample predictive accuracy of the algorithm is not dependent on one particular measure of director performance by considering two alternative outcome measures in addition to excess votes. The first is a dichotomous variable (*Dissent*) that captures low absolute shareholder support, that is, strong dissent against a director. The second is based on the idea that a director who leaves shortly after being appointed likely reflects a poor director-firm match (Ferreira et al. 2017, Bates et al. 2017). We therefore use a dichotomous variable (*Director turnover*) for turnover within 2 years of appointment as an alternative performance measure.⁴ In unreported tests, we also use total instead of excess votes to measure director performance and draw similar conclusions. Finally, we consider the model's ability to predict announcement returns of director appointments. We find that the algorithm's predictions of shareholder votes to reelect directors are also strongly related to announcement returns around director appointments.

We observe director performance for directors who were nominated to the board, but not for potential candidates who were not nominated. This *selective labels* problem of observing director performance only for directors who were actually selected is a form of selection that is common in prediction problems (see Kleinberg et al. 2017). If boards are skilled at using

⁴ Our results are unchanged if we use turnover within 3 years instead of 2 years.

unobservables in their nominating decisions, nominated directors could have higher expected performance than otherwise similar (based on observables) passed-over directors. To examine this issue, for each board appointment in our test set, we construct a realistic pool of potential candidates: directors who joined the board of a smaller neighboring company within a year. Although we do not observe the outcome (the *label*) of those potential candidates at the focal firm, the design of our candidate pools allows us to observe what we refer to as their *quasi-label*: their observed performance on the board they effectively joined. We use the distribution of quasi-labels as a substitute for the distribution of missing labels we would have observed for passed-over directors at the focal firm. This approach allows us to approximate the quality of alternative candidates who could have been selected and assess whether the algorithm's predictions are informative about how a director will perform relative to these alternatives: we test whether the percentile rank (relative to alternative available candidates) is worse for focal directors the algorithm predicted would do poorly than for those it predicted would do well.

We find that directors the algorithm predicted would perform poorly (well) do perform poorly (well) when compared to potential available alternatives. The average director in the bottom decile of predicted performance ranks at the 23rd percentile in the distribution of quasi-labels. In contrast, those in the top decile rank at the 80th percentile. The difference is statistically significant. OLS models are unable to predict *ex ante* who will perform well compared to alternatives and who will not.

One of the differences between machine learning algorithms and traditional econometric modeling is that machine learning algorithms do not provide a formula that can be used to infer the influence of any particular independent variable on the dependent variable. However, in recent years, rapid developments in the *Explainable AI (XAI)* or *Interpretable ML* strand of the machine learning literature have focused on improving model interpretability (e.g., Lundberg and Lee 2017; Ribeiro, Singh, and Guestrin 2016; for a review, see Vilone and Lungo 2020). We employ methods from this literature to gain insights into our algorithm and quantify the contribution of each feature to predicting director performance.

In addition, while machine learning models do not generate estimates of the underlying structural parameters of a model, we can use the algorithm's predictions to understand the features that are overvalued and undervalued by firms in the director selection process. Relative to algorithm-selected directors, board-selected directors who receive predictably low shareholder approval are more likely to be male, have larger networks, and sit on more boards. These attributes characterize the stereotypical director in most large companies. A plausible interpretation of our results is that firms that nominate predictably unpopular directors tend to be subject to homophily when choosing directors, while the algorithm suggests that adding diversity would be beneficial.

Finally, to help understand the process determining the nomination of directors, we attempt to distinguish between potential explanations (error or agency) for why firms regularly make predictably poor decisions when they hire directors. We find that firms that nominate directors who were predictably poor choices tend to have governance structures that disenfranchise shareholders. This pattern is consistent with the view that firms choosing predictably bad directors is a manifestation of underlying agency conflicts within a boardroom.

Several papers in the recent economics and finance literature have used machine learning techniques. In a seminal paper, Kleinberg et al. (2017) study judges' bail decisions and show that machine predictions could significantly reduce crime. Machine learning is quickly being adopted as a new methodology in the asset pricing literature (e.g., Rossi 2018; Ke, Kelly, and Xiu 2019; Abis 2017 among others) and in microstructure (e.g., Easley et al. 2020). Corporate finance applications are developing (see, e.g., Li et al. 2020 and Bubb and Catan 2020). Our paper uses machine learning to add to the literature on corporate governance, and in particular on the selection of directors (Smith 1776; Berle and Means 1932; Hermalin and Weisbach 1998; Kramarz and Thesmar 2013; Coles, Daniel, and Naveen 2014; Cai, Nguyen, and Walkling 2017).

1. Constructing a Sample on which Algorithms Can Select Directors

1.1 Measuring director quality

An essential part of designing the algorithm is specifying a measure of director performance as the basis for which directors are selected. Our analysis focuses on the relative shareholder support that directors receive in annual director reelections as a market-based measure of *individual* director performance. Our main outcome variable is *Excess votes*: the average level of shareholder support over the first 3 years of director tenure, adjusted by the average support for the entire slate of directors up for reelection on that board that year.⁵ Our results are qualitatively unchanged if we task the algorithms with predicting the total level of shareholder support, that is, if we do not subtract the average for the slate.

One potential concern with using shareholder support as our measure of director performance is that in the vast majority of cases, all directors receive an overwhelming majority of the votes, with mean shareholder support usually around 95%.⁶ There is almost no variation in the *outcome* of the reelections. Nonetheless, variation among winning votes does appear to reflect differences

⁵ The distribution of shareholder support does not change over the first few years of a director's tenure. We obtain similar results using shareholder support at year 1, year 2, or year 3 instead of using the average over the first 3 years.

⁶ The literature on director reelections is large, including Boone, Field, and Karpoff (2007), Linck, Netter, and Yang (2008, 2009), Cai, Garner, and Walkling (2009), Fischer et al. (2009), Coles, Daniel, and Naveen (2014), Iliev, Lins, Miller and Roth (2015), Aggarwal, Dahiya, and Prabhala (2017), Ertimur, Ferri, and Oesch (2018), Cai, Nguyen, and Walkling (2017), Fedaseyev, Linck, and Wagner (2018), Fos, Li, and Tsoutsoura (2018).

in directors' quality. Cai et al. (2009), Fischer et al. (2009), and Iliev et al. (2015) find that vote totals predict stock price reactions to subsequent turnover. In addition, vote totals are negatively related to CEO turnover, board turnover, management compensation levels, as well as the probabilities of removing poison pills and classified boards.

Moreover, director reelections appear to affect a firm's real activities, even if the elections are not contested and the nominated directors are reelected. Fos, Li, and Tsoutsoura (2018) find that when directors are closer to getting reelected, firms are more likely to fire CEOs, presumably to persuade shareholders of the firm's diligence. Aggarwal, Dahiya, and Prabhala (2017) suggest that directors with low relative support are more likely to leave the board, and if they stay, tend to move to less prominent positions. And Ertimur et al. (2018) find that when votes are withheld from directors, boards explicitly attempt to address shareholders' concerns.

Shareholder support could reflect recommendations by proxy advisors, such as ISS. Ertimur et al. (2018) report that since 2003, large institutional investors take an active role in developing the guidelines that are the basis of ISS recommendations, which, as such, reflect its clients' aggregated preferences. Aggarwal, Erel, and Starks (2016) confirm this result, documenting that institutional investors and proxy advisors pay attention to the changing opinions of their beneficiaries and shareholders. However, institutional investors do *not* follow proxy advisors' recommendations blindly. Aggarwal, Erel, and Starks (2016) find that shareholders are less likely to follow the recommendations of either management or proxy advisory firms as shareholders are forming their own views following changes in public opinion. Iliev and Lowry (2014) show that institutional investors with larger size of ownership tend to vote more independently from ISS recommendations.

We therefore repeat our tests by focusing on a subsample of firms with larger-than-median ownership ($>26\%$) by the top-five institutional owners and our results are unchanged. Using detailed voting data from 2003 to 2017, Heath et al. (2019) show that when ISS recommends voting against management, index (active) funds vote with management 54% (42%) of the time. This recent stream of the literature strongly suggests that shareholder votes are not simply the reflection of recommendations issued by proxy advisors.

Overall, the literature finds that the level of shareholder support does reflect perceptions of director quality, that directors do care about these perceptions, and take actions to influence them. We test whether algorithms can pick up variations in these perceptions of director quality despite the fact that most directors receive extremely high support.

Shareholders do on occasion oppose newly nominated directors. Therefore, we also train the algorithm to predict strong dissent, measured by an indicator variable equal to one if a director receives low (less than 90%) support, as an alternative measure of director performance using shareholder votes.

Finally, director turnover is another measure of director-firm match quality used in the literature (see, e.g., Ferreira et al. 2017). Therefore, we also use whether a new director leaves within 2 (or 3) years of his or her appointment as an alternative measure of director quality.

1.2 Sample selection

To evaluate the performance of an algorithm to select directors, we must gather a sample in which we can observe the attributes of firms and boards, and also measure the performance of directors. Because of these requirements, we focus on a sample of boards from large, publicly traded, U.S. firms with an average market capitalization of \$6.6 billion. We identify 41,015 new independent directors appointed to 4,887 unique corporate boards of these firms between 2000 and 2014 using *BoardEx*, which is our main data source for director and board-level characteristics. Internet Appendix IA1 defines the variables in detail.

We obtain data on the level of shareholder support for individual directors from *ISS Voting Analytics* and focus on directors appointed during our sample period. Because of the possibility of factors that lead all directors in one firm to receive higher average votes from directors in other firms, we rely in most specifications on a measure of *Excess votes*. To construct this measure, we start with the number of votes in favor over all votes cast (yes, no, withheld). We then subtract the average for the slate of directors up for reelection on that board, and take the average of this variable over the first 3 years of tenure.⁷ Our sample contains the voting outcome, *Excess votes*, for 24,054 new director appointments.⁸

1.3 Summary statistics

Table 1 presents summary statistics for the four measures of director performance. As previously documented in the literature on uncontested director elections, the overall level of shareholder support is typically very high. Given that the mean level of support is 0.948 and the median is 0.975 (with a standard deviation of 0.07), a voting outcome below 90% is a relatively poor outcome for a director. Therefore, we also present the means for strong dissent against a director, which is a dummy variable that takes on a value of one if the director receives less than 90% of shareholder support, as well as a dummy for director turnover within 2 years of appointment.

Table 2 illustrates that the frequency of shareholder discontent varies by director and board characteristics. For example, the fraction “poor outcomes,”

⁷ Some firms have “staggered” boards, which are elected for 3-year terms. For these firms, instead of averaging the support over the first three elections of a director’s tenure, we use the support in the one election covering the 3-year period.

⁸ All the results reported below are similar when we use shareholder support without adjusting for the average support of the other directors at the firm.

Table 1
Summary statistics for director performance measures

	N	Mean total votes	Mean excess votes	Mean dissent	Mean turnover
2000	331	0.950	0.001	0.127	0.132
2001	772	0.944	0.000	0.134	0.131
2002	1,057	0.946	0.002	0.118	0.086
2003	1,774	0.951	0.006	0.087	0.081
2004	2,019	0.953	0.007	0.086	0.104
2005	1,893	0.948	0.005	0.103	0.108
2006	1,789	0.941	0.005	0.129	0.099
2007	1,942	0.940	0.005	0.160	0.095
2008	1,691	0.944	0.007	0.155	0.112
2009	1,541	0.948	0.007	0.116	0.111
2010	1,842	0.948	0.004	0.136	0.114
2011	1,825	0.954	0.004	0.118	0.114
2012	1,862	0.952	0.005	0.111	0.113
2013	2,148	0.948	0.003	0.131	0.126
2014	1,568	0.959	0.006	0.094	0.098
	24,054	0.948	0.004	0.120	0.108

This table presents the *mean* for total and excess shareholder support over time, as well as for strong dissent against a director and director turnover. Shareholder support is defined as the fraction of votes in favor of a given director over all votes cast for the director’s reelection within 3 years of her tenure. Dissent is equal to one for a director whose average total votes is less than 90% within 3 years of her tenure. To compute *Excess votes*, we subtract the average of that variable for the slate of directors up for reelection that year on the focal board. We take the average of this relative vote measure over the first 3 years of the new director’s tenure. Turnover is equal to one if the director leaves the board within 2 years of being appointed. The data come from ISS Voting Analytics and BoardEx.

Table 2
Average fraction of poor outcomes

	Full sample	yes	no	Difference p-value
Director level				
Male	0.102	0.106	0.079	0.000
Foreign	0.101	0.115	0.100	0.138
Qualifications > median	0.102	0.094	0.106	0.005
Network size > median	0.102	0.108	0.096	0.002
Generation BBB	0.101	0.093	0.118	0.000
Generation X	0.101	0.151	0.096	0.000
Busy director	0.102	0.145	0.090	0.000
Finance background	0.102	0.106	0.101	0.328
Board level				
Fraction male > median	0.102	0.116	0.091	0.000
Board size > median	0.102	0.089	0.114	0.000
Nationality mix > median	0.102	0.108	0.100	0.064
Attrition rate > median	0.098	0.106	0.086	0.000

This table presents the average fraction of “poor outcome” for various director- and board-level characteristics. A director is considered to experience a poor outcome if her excess votes is < −2 %. Poor outcomes represent 10% of the sample.

representing the bottom 10% of the sample in terms of excess votes, is 10.6% for male directors and 7.9% for female directors.⁹ Similarly, busy directors

⁹ The pattern is similar if we define a poor outcome as having shareholder support below 80%.

(serving on three or more boards) experience low shareholder support more frequently than nonbusy directors.¹⁰

Although some variables appear to affect director performance, theory provides little guidance regarding the particular variables and functional forms of the relation between the various director, board and firm characteristics and the performance of directors. For example, we do not know whether we should expect busy female directors with a PhD serving on the large board of a small firm in the pharmaceutical industry to receive higher or lower shareholder support on average than a male director who serves on a single small board of a large manufacturing corporation. The problem increases in complexity when many more covariates are likely to matter. For this reason, we wish to utilize an estimation procedure that does not impose the specific form for the relationship between potential explanatory variables.

This logic is one reason machine learning algorithms are successful at prediction. They are designed for problems, such as predicting which directors will be successful in a given firm, on which theory is silent on the appropriate functional form between the explanatory variables and the outcome to predict and allow for rigorous, data-driven model selection (Athey 2017).

2. Evaluating Machine Learning Predictions of Director Performance

2.1 Model specification

We employ machine learning algorithms that predict the performance of potential directors. The algorithms employ a set of observable director, board, and firm features (see Internet Appendix IA1) that is available to the nominating committee at the time of the nominating decision. The algorithms are commonly used in the supervised machine learning literature: *Lasso*, *Ridge*, *Neural Networks*, and *Gradient Boosting Trees (XGBoost)*.¹¹ We first train each algorithm on the 2000–2011 portion of our sample containing 18,476 new independent director appointments, of which 12,815 are unique directors, at 2,407 firms. Training involves having the algorithm determine which combinations of variables best predict future performance.¹² We evaluate the models' out-of-sample predictions on the held out 2012–2014 portion of our sample containing 5,578 new director appointments, of which 4,019 are unique directors, at 569 firms. We compare these out-of-sample predictions to those from an AIC-selected OLS model (see Table C.1 in Appendix C). All comparisons are based on predictions for the 2012–2014 subsample of director

¹⁰ Fich and Shivdasani (2006) present evidence suggesting that a director being overly busy can meaningfully affect their monitoring of management.

¹¹ *XGBoost* is a scalable ensemble tree-based method developed by Chen and Guestrin (2016) and can be easily utilized on a standard laptop using the *XGBoost* package for Python, available at <https://pypi.org/project/xgboost/>.

¹² The algorithms rely on a regularizer that balances in-sample fit versus out-of-sample overfitting.

Table 3
Using OLS versus machine learning to predict director performance

	Predicted Percentile of Excess Votes	OLS	XGBoost	Ridge	Lasso	Neural Network
Directors predicted to perform	1%	0.030	−0.031	−0.017	−0.026	−0.017
	< 5%	−0.002	−0.012	−0.011	−0.018	−0.011
poorly	10%	0.007	0.001	−0.004	0.002	−0.003
Directors predicted to perform well	90%	0.004	0.008	0.012	0.004	0.013
	< 95%	−0.004	0.016	0.017	0.010	0.011
	100%	0.006	0.012	0.012	0.018	0.014

This table reports the average observed level of excess shareholder support over the first 3 years of a new director’s tenure for directors who were ranked by their predicted level of shareholder support by an OLS model and several machine learning algorithms (*XGBoost*, *Ridge*, *Lasso*, and *Neural Network*). Shareholder support is defined as the fraction of votes in favor of a given director over all votes cast for the director’s reelection within 3 years of her tenure. To compute *Excess votes*, we subtract the average of that variable for the slate of directors up for reelection that year on the focal board. Then we take the average of this relative vote measure over the first 3 years of the new director’s tenure.

appointments, which does not overlap with the 2000–2011 subsample on which the algorithms are trained.

The optimal way to choose the size of the training and test sets depends on the signal-to-noise ratio in the data and the training sample size. Therefore, establishing a general rule on how much training data are enough is difficult to do. For very large data sets, a 90%/10% split can be done, although 70%/30% or 80%/20% splits are typically used in practice.¹³ We use an 80%/20% split but our results do not depend on the way in which we split the data into training and test periods.¹⁴ Because of the possibility that the Sarbanes-Oxley Act (SOX) affected the process by which independent directors are chosen, we have also used a training sample starting in 2003 (keeping the test sample between 2012 and 2014). In Figure IA2 in the Internet Appendix, we report results, which are very similar to those discussed below, using this training sample.

2.2 Predictions of director performance

A way to evaluate the quality of a model predicting performance is to evaluate the way in which actual performance increases related to predicted performance. Table 3 summarizes the ability of the machine learning models, once trained on the earlier portion of the sample, to predict director success in the later part. Table 3 indicates that average observed shareholder support increases across model-predicted performance percentiles for each machine learning model. In contrast, in the OLS model, there is no relation between predicted and actual director performances.

¹³ See Hastie et al. (2009) for a discussion of methodological issues involved in choosing training and testing sets.

¹⁴ Our results are similar when we include appointments from 2011 to 2014 in the test set (see, e.g., Internet Appendix Figure IA1).

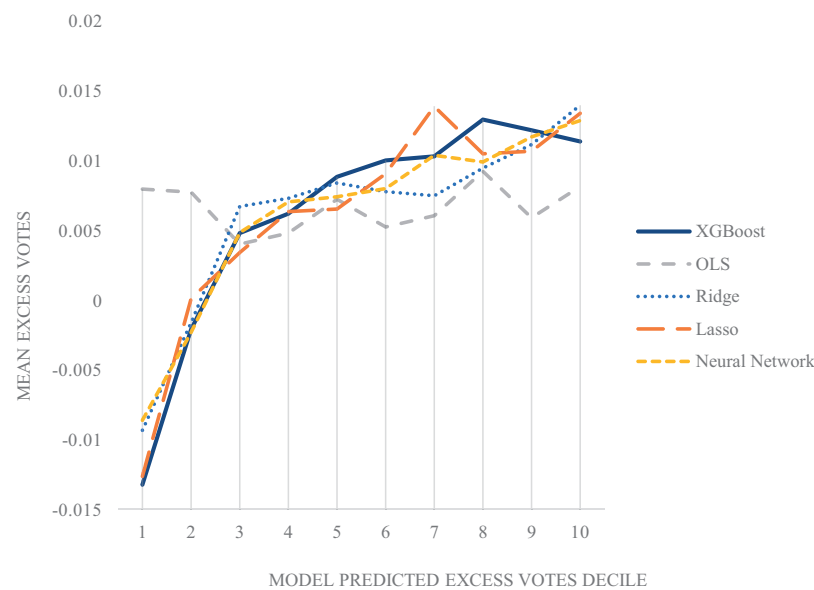


Figure 1
Mean observed *Excess votes* versus predicted *Excess votes*

This figure shows the average observed level of excess shareholder support for directors across the 10 deciles of predicted performance for OLS and ML models in the 2012–2014 test set. To compute *Excess votes*, we first compute the fraction of votes in favor of a given director over all votes cast for the director. Next, we subtract the average of that variable for the slate of directors up for reelection that year on the focal board. Finally, we take the average of this relative vote measure over the first 3 years of the new director’s tenure.

Among the machine learning algorithms, *XGBoost* and *Lasso* perform best at predicting the subsequent success of directors using excess votes. Directors predicted to be in the bottom percentile as predicted by *XGBoost* and *Lasso* have an average *observed* excess shareholder support of -3.1% and -2.6% , respectively, whereas the average observed excess support is 1.2% and 1.8% for directors in the top percentile of predicted performance.

Figure 1 presents the average observed level of shareholder support for directors across 10 deciles of predicted performance for OLS and for the machine learning algorithms in the 2012–2014 test period. The figure documents that the mean shareholder support for a director is an increasing function of predicted support for all the machine learning algorithms, but not for the OLS model.

The inability of the OLS model to predict director performance could potentially occur because the particular model we picked was not well specified. For this reason, in Table C.1 in Appendix C, we present various OLS specifications that include director, firm, and board-level variables that have been used in the literature. We also include industry and time fixed effects in various specifications. It is important for us to use only the *ex ante* variables that

would be available to the nominating committee when they pick new directors.¹⁵ For each model, we present specifications with *Excess votes* as the dependent variable (with and without year and industry fixed effects) as well as with *Total votes* (including firm-year fixed effects; see Gormley and Matsa 2014). To compare the out-of-sample predictable power of these specifications, we calculate the AIC for each model. The AIC provides each specification's out-of-sample prediction error and allows us to compare the relative quality of OLS models presented. The OLS model used in Table 3 corresponds to Model (5) of Table C.1 in Appendix C, which has the lowest AIC error.

3. Comparing the Performance of Directors to that of Potential Alternative Choices

Our results so far document that directors identified by our algorithm as likely to have low (high) future shareholder support, are in fact more likely to receive low (high) support in subsequent elections.¹⁶ The key issue raised by this finding is that when firms hire directors who reveal themselves to be poor hires, it is possible, at least some of the time, to predict this poor performance before the choice was made. In other words, firms regularly hire *predictably poor* directors.

One possible explanation for this practice is that there are simply no better alternatives available to the firms, and although it is predictable that these directors will do poorly, others whom the firm could have hired would have been even worse. This possibility illustrates a form of selection bias referred to in the machine learning literature as the “selective labels problem” (Kleinberg et al. 2017). This problem refers to the fact that only the firms' choices for directors that actually occurred are observed, so it is impossible to know how potential alternative directors who they did not select would have performed. This possibility may limit our ability to assess how algorithmic decision aids could improve on boards' decisions.

Our setting allows us to evaluate the importance of this issue. We consider a sample of individuals who would have taken the directorship with high probability. We classify these alternative candidates as those directors who, within 1 year of the appointment, joined the board of a smaller neighboring firm.¹⁷ These directors were available to join a board at that time and were willing to travel to that specific location for board meetings. We restrict the

¹⁵ Cai, Garner, and Walkling (2009) examine the determinants of shareholder votes at annual director elections using an OLS regression. The R^2 of the model drops significantly when explaining votes for new directors only and when removing variables not available to board members at the time of hiring (e.g., ISS recommendation).

¹⁶ In an earlier version of this paper, we ensure that our results are not driven by poorly performing firms. Excluding firms that had negative abnormal returns in the year prior to the nomination produces results similar to those reported here. See Erel et al. (2018).

¹⁷ A neighboring firm is defined as a firm whose headquarters is within 100 miles of the focal firm's headquarters.

pool of potential candidates to directors who joined a smaller neighboring firm since the prestige and remuneration of being a director tends to increase with company size (see Masulis and Mobbs 2014). There are on average 147 candidates in a candidate pool.¹⁸

While we do not observe the performance of these potential candidates at the focal firm, we do observe their performance on the board they did join. In the absence of a direct measure of performance at the focal firm, this measure represents an informative signal, which we refer to as a potential director's *quasi-label*. We do not rely on the assumption that one particular director's performance at firm B would be the same as her performance at firm A. Instead, for each candidate pool associated to a focal director, we use the distribution of quasi-labels to approximate the performance distribution that these candidates would have had on the focal board. In other words, this distribution of quasi-labels captures the quality of alternative candidates the focal board could have considered.

For each decile of predicted performance for focal directors, we estimate the average (median) percentile rank of the focal director's observed performance in the distribution of quasi-labels, and examine whether this rank is correlated with predicted performance. The null is that a director's percentile rank in the distribution of quasi-labels is uncorrelated with her predicted performance; that is, the prediction is not informative of future performance when compared to other candidates. The alternative is that the percentile rank for directors predicted to perform poorly (well) by the algorithm is lower (higher) than that for those predicted to do well (poorly). Appendix B.2 provides a simple formalization of how we evaluate algorithmic predictions using quasi-labels and details the assumptions underlying the approach. The procedure can be summarized as follows:

1. Identify the peers $\{-i\}$ of each director i as discussed above.
2. Compute the percentile rank r_i of the focal director's actual vote y_i in the distribution of the peer director votes $\tilde{y}_{-i}$.
3. Test whether the percentile rank r_i is higher for the directors the algorithm predicts will do better.

The actual implementation sorts directors into 10 deciles based on the algorithm-predicted votes $\hat{y}_i$. It then plots the average rank $\bar{r}_i$ for each decile and finds a monotonic relationship. Thus, the average better-ML-predicted director does well not only in an absolute sense but also relative to peers.

Consider directors with low predicted performance. Intuitively, if their observed performance were to lie in the right tail of quasi-labels, this would

¹⁸ We redo our analyses and relax the assumption that candidates join the board of a smaller firm and find that the results remain robust. We report the results using this larger pool of candidates in Internet Appendix Figure IA3. Our results also hold when we restrict pools to candidates who joined the board of a company in the same industry as the focal firm.

Table 4
Evaluating the predictions using quasi-labels

	OLS	XGBoost	Ridge	Lasso	Neural Network
Bottom decile of predicted performance	66 th	23 rd	33 rd	25 th	35 th
Top decile of predicted performance	70 th	80 th	80 th	84 th	77 th

This table reports how nominated directors rank in the distribution of quasi-labels of their candidate pool. For each nominated director in our test set, we construct a pool of potential candidates who could have been considered for the position. Those candidates are directors who accepted to serve on the board of a smaller nearby company within a year before or after the nominated director was appointed. The quasi-label for each of these candidates is how she performed on the competing board she chose to sit on. The first (second) row shows the median percentile of observed performance in the distribution of quasi-labels for directors the model predicted to be in the bottom (top) decile of predicted performance. Each column presents the results from a different model.

imply that they ended up doing well relative to available alternatives, even though our algorithm had predicted they would do poorly. The focal board could have selected on unobservables, and the high rank in the distribution of quasi-labels would suggest that the unobservables were used as *signal*. It could also suggest that supply considerations were such that there was a shortage of good candidates and even though the director did not do well, alternative candidates did even worse. On the other hand, if their observed performance were to lie in the left tail of quasi-labels, this would imply that our algorithm identified *ex ante* that these directors would perform poorly, and relative to alternatives, they indeed did perform poorly. This pattern would suggest that there were better alternative candidates and that any unobservables used in the nomination decision process were not a signal of performance, but instead captured noise, bias, or agency problems.

We evaluate this procedure in two ways. In the first, the algorithm generates performance predictions for potential candidates on the focal boards and we use these predictions to narrow down the candidate pools. For focal directors predicted to do poorly (well), we narrow down the candidate pool to those the algorithm predicted were most (least) promising. Table 4 presents the median rank in the distribution of quasi-labels for directors in the bottom and top deciles of predicted performance for several machine learning algorithms, as well as for the OLS model. For all machine learning models, nominated directors predicted to do poorly performed noticeably worse than available promising candidates, while nominated directors predicted to do well performed better than available unpromising alternative candidates. *XGBoost* and *Lasso* again appear to be the preferred algorithms as they can best discriminate the directors who will do well from those who will not. In contrast, the predictions from the OLS model are uninformative about subsequent performance relative to available alternative candidates.

In a second test of this procedure, we compare the performance of focal directors to the performance of *all* potential candidates and present this comparison in Figure 2. Using the predictions from the *XGBoost* algorithm, the mean and median percentile rank in the distribution of quasi-labels increase

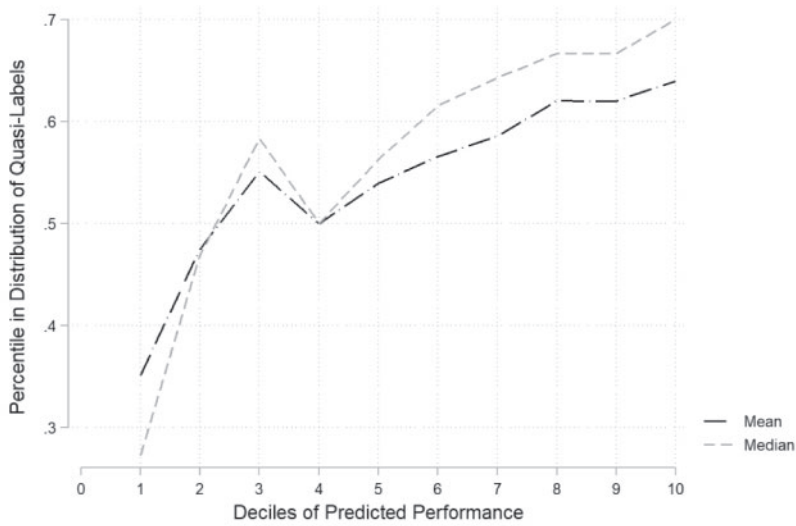


Figure 2
Mean and median ranks in quasi-label distribution across deciles of predicted performance

This figure shows the mean and median ranks in the distribution of quasi-labels for directors in each of the 10 deciles of *XGBoost*-predicted performance (*Excess votes*). The observed performance of nominated directors in our test set is compared to the quasi-labels of *all* potential candidates in their respective candidate pool: each new board appointment in the test set is associated with a candidate pool, comprising directors who, within 1 year of the appointment, joined the board of a smaller neighboring firm.

almost monotonically across deciles of *predicted* performance. This suggests that as the algorithm’s prediction of a director’s performance increases, her effective percentile rank when compared to plausible candidates increases. The difference between decile 1 and 10 is statistically significant at the 1% level (both for the mean and the median).¹⁹

The goal of this section was to evaluate algorithmic predictions through an approach that considers the performance of alternative choices of directors. The results provide additional evidence that algorithmic predictions are useful and have potential to improve nominating decisions.

4. Alternative Measures of Performance

An important concern is the extent to which, being trained to predict *Excess votes*, the algorithm is actually predicting a director’s performance, or merely predicting which directors will be popular with shareholders. Hart and Zingales (2017) argue that a director’s performance is definitionally equal to her

¹⁹ In the rest of the paper, to save space, we report results using *XGBoost* in the main text and the ones using *Lasso* in Appendix A. Although *Lasso* performs well at predicting the distribution of outcomes in this exercise, *XGBoost* provides better predictions using alternative definitions of director performance.

popularity with shareholders. However, others would disagree, claiming that a director's performance is her impact on a firm's profitability regardless of what shareholders say about her.

One reason this issue is a concern is that many institutional shareholders decide on their votes through recommendations of shareholder services companies, such as ISS. ISS introduced guidelines in the latter part of our training period. For example, explicit guidelines to support proposals aimed at increasing female board representation were introduced in 2010.²⁰ As discussed before, we find similar results when we focus on a subsample of firms with larger-than-median ownership by the top-five institutional owners (Iliev and Lowry 2014), who tend to rely on ISS recommendations to a smaller extent.

4.1 Predicting abnormal returns around director appointment announcements

The recent literature on routine director reelections does find that votes capture the performance of directors. We confirm that this pattern occurs in our data as well. We compare the cumulative abnormal returns (CARs) around the announcement of director appointments in our test set for directors predicted to do well to those for directors predicted to do poorly.²¹

Table 5 reports the mean CARs using a $(-1, +1)$ window around announcements. Using *XGBoost* to predict excess votes, we find that the mean CAR for directors predicted to do poorly (decile 1) in our test set is -1.94% whereas it is $+0.75\%$ for directors predicted to do well (decile 10). The difference is statistically significant at the 1% level. Directors predicted to be unpopular also tend to be viewed by the market as worse directors. We also used the algorithm to predict announcement CARs using a smaller sample for which announcement dates are available, and also with larger event windows, with similar results.²²

4.2 Predicting strong dissent against a new director

It is possible that only when dissent is particularly strong is it a useful signal of director performance. With this notion in mind, we create a dummy variable that equals one if there are more than 10% dissenting votes in their reelection (within the next 3 years) from shareholders, and use it as an alternative measure of poor director quality. Since the mean shareholder support in our sample is

²⁰ Our training sample covers data from 2000 to 2011. Less than 20% of appointments in our training set take place when ISS had those specific guidelines in place.

²¹ We collect announcement dates from *BoardEx*, *CapitalIQ*, and *Lexis-Nexis*.

²² An alternative approach to looking at the stock price reaction to the announcement of the director's appointment is to examine whether the predicted quality of a director is associated with changes in the firm's profitability. An earlier draft of this paper performed this analysis and concludes that excess votes are closely related to profitability, so that if the model is trained on excess shareholder votes, it also predicts profitability and vice versa. See Erel et al. (2018).

Table 5
Cumulative abnormal returns around appointment announcements

	N	Mean	Median
Directors in Decile 1 of predicted performance (excess votes)	292	−1.94%	−0.64%
Directors in Decile 10 of predicted performance (excess votes)	575	0.75%	0.34%
Difference in means (p-value)		0.0043	

This table reports the mean and median cumulative abnormal returns for directors predicted to do poorly and for directors predicted to do well. Directors predicted to do poorly (well) are directors in decile 1 (decile 10) of predicted performance (excess votes) as predicted by the *XGBoost* algorithm. Results are shown for appointments in the test set only. The cumulative abnormal returns reported are computed using a (−1, +1) window.

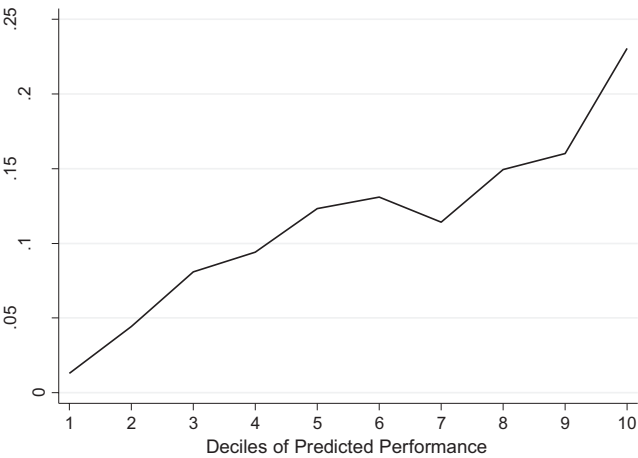
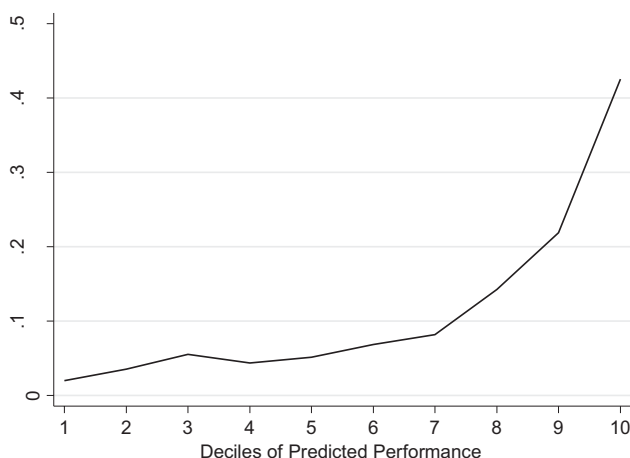


Figure 3
Mean observed Dissent versus predicted Dissent

This figure shows the average observed level of dissent against a director by shareholder (the fraction of new directors who received less than 90% shareholder support in their reelection within the next 3 years) across the 10 deciles of *XGBoost*-predicted dissent in the 2012–2014 test set. Directors in decile 1 (10) were predicted by *XGBoost* as least (most) likely to face strong dissent. The unconditional mean dissent is about 11%.

95% for a new director in his/her reelection, we consider a support level of less than least 90% to be a particularly bad outcome.

In Figure 3, we repeat our exercise of predicting director performance using the training set of 2000–2011 and comparing it with the actual performance in the test set of 2012–2014, this time using whether the director receives less than 90% shareholder support as our measure of performance. We find that only 1.3% of directors in the bottom decile of predicted dissent (i.e., directors the algorithm predicted had the lowest chance of receiving less than 90% support) face strong dissent from shareholders. This ratio increases to 23% for directors in the top decile of predicted dissent.

**Figure 4****Mean observed *Director turnover* versus predicted *Turnover***

This figure shows the average observed level of director turnover (the fraction of new directors who left within the next 2 years) across the 10 deciles of *XGBoost*-predicted turnover in the 2012–2014 test set. The unconditional mean turnover is about 11%.

4.3 Predicting director turnover

Director turnover shortly after being appointed is likely to be associated with a poor fit between the director and the firm, suggesting that the decision to hire that director was misguided. For this reason, we train the *XGBoost* algorithm to predict whether a director will leave within 2 years following her appointment. We choose the 2-year span to ensure that the turnover happens before reelection; however, the results are similar if we use turnover over a 3-year period.

Figure 4 documents the way in which the *XGBoost* algorithm's predictions compare with actual director turnover. The figure shows the average observed director turnover within 2 years of appointment across the 10 deciles of *XGBoost* predicted turnover in the 2012–2014 test period. Figure 1 documents a clear monotonically increasing relationship between the average fraction of directors who departed within 2 years in the test sample and the algorithm's predictions. The difference in mean turnover between the bottom and top deciles is large in magnitude: while less than 2% of directors in the bottom decile leave within 2 years, this fraction increases to 43% in the top decile. The unconditional mean is about 11%.

The estimates using these alternative measures of director performance are consistent with the ones presented above using *Excess votes*. Regardless of the measure we use, the algorithm can predict the distribution of future director performance.

5. Characteristics that Affect Director Performance

One of the differences between machine learning algorithms and traditional econometric modeling is that machine learning algorithms do not provide a

formula that can be used to infer the influence of any particular independent variable on performance. With the goal of understanding what leads models to make specific predictions, the machine learning literature has focused on developing methods to improve model interpretability and to make black-box models more transparent (e.g., Lundberg and Lee 2017; Ribeiro, Singh, and Guestrin 2016).

In this section, we employ one such method, SHAP (Shapley Additive exPlanations), to understand which variables are most important in our model's assessment of the quality of a potential director. SHAP values were introduced by Lundberg and Lee (2017) as a game theoretic approach to improve the interpretability of a complex model's output. A feature x 's SHAP value estimates the extent to which it contributes to pushing the model's output away from the base value (the unconditional expectation). It is defined as the change in the expected model output, averaged across all possible orderings of the other (non- x) features. Each feature's incremental contribution to the model's output is estimated by considering the output of all possible combinations of features. This analysis provides a way to identify the variables that are especially important when training the algorithm.²³

5.1 Factors affecting the predictions of director quality

We would like to use the algorithm's predictions to learn more about the decision-making process that governs the nomination of corporate directors. For that purpose, we examine the importance of characteristics that our algorithm uses to predict a given director's performance. We use SHAP values to quantify the marginal contribution of each feature to predicting director performance. A feature's SHAP values can be computed for each observation, that is, for each individual prediction, to improve the model's transparency (*local interpretability*). This approach is helpful as it shows which variables were instrumental in generating a specific prediction. Additionally, SHAP values for each feature can be averaged across observations. This analysis produces a ranking of variables to understand which most contribute to the model's output (average across observations), and therefore improves the *global interpretability* of the model.

5.1.1 Global interpretability. Figure 5 presents the 10 features that contribute the most to *XGBoost*'s predictions of director performance, measured by SHAP values averaged across observations. Panel A uses excess votes as the performance measure, with the variables presented in red representing the ones that contribute positively to predicting director success, while the variables presented in blue representing those that contribute negatively to success. For

²³ For this analysis, we use the SHAP library for Python available at <https://github.com/slundberg/shap>. The task of estimating SHAP values is computationally complex. We refer readers interested in the details of this computation to Lundberg and Lee (2017).

example, being a member of the compensation committee decreases the excess vote prediction by 0.22% on average, while being on the audit committee increases it on average by 0.16%. The standard deviation of incumbent directors' time on board increases the predicted performance of the incoming directors, while the total number of boards she sits on currently and sat on in the past decreases her predicted performance.

In panel B, we calculate SHAP values when the model predicts whether a director receives less than 90% of the votes. This panel indicates that the attributes with the largest SHAP values when predicting excess votes largely overlap with those with the largest SHAP values when the model predicts dissent.

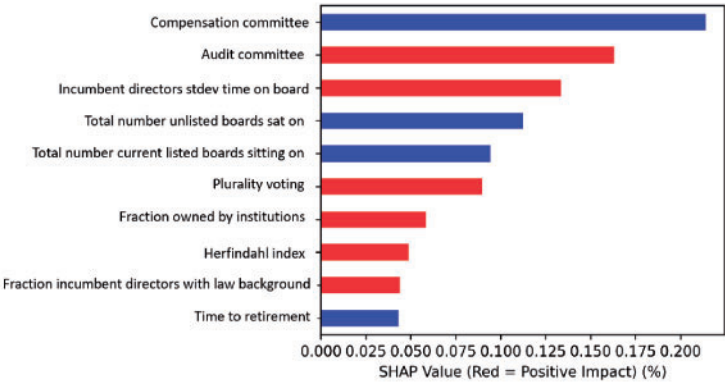
In panel C, we present SHAP values for models that predict director turnover within 2 years of joining the board. Mostly individual director characteristics appear to explain turnover predictions. For example, being the chairman, having an entrepreneurial background, or having graduated from an Ivy League university. Having a classified board, which gives directors longer term, has a negative effect on board turnover predictions.

We emphasize that SHAP values do not establish causality. Instead, they quantify features' contribution to the model based on correlations. However, SHAP values do provide a helpful basis for expanding our qualitative understanding of how machine learning models generate predictions.

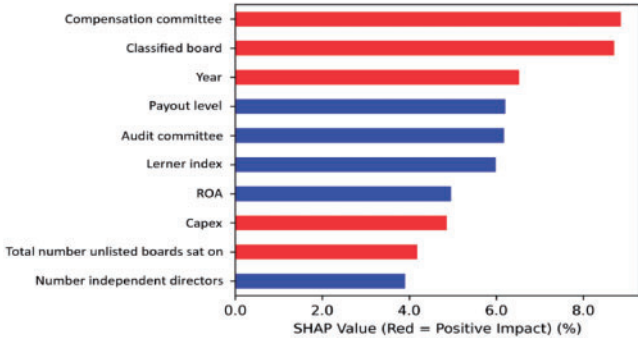
5.1.2 Local interpretability. We can also compute a feature's SHAP values for individual observations. Figure 6 reports the features that most contributed to three randomly selected observations, one for each measure of director performance. For each, the "model output value" is the model's prediction for this specific observation ($\hat{y}_i$), while the "base value" is the mean prediction across all observations. Features in red (blue) increase (decrease) the model's outcome, that is, the individual prediction $\hat{y}_i$. The value of the attribute (x) is reported for each observation. The arrow's length for each attribute corresponds to its SHAP value (i.e., longer arrows represent more important attributes for this observation). For each observation, the difference between the model's output and the base value (mean prediction) is the sum of all features' SHAP values.

For example, in panel A, where the model predicts excess votes, the most important variable affecting the model's prediction is that the incoming director sat on a total of eleven unlisted boards, which pushed the predicted *Excess votes* downward. However, the fact that the director is joining the audit committee, but not the compensation committee, pushed the model's performance prediction upward. The standard deviation of incumbent directors' time on the board is 6 years (i.e., 2 years above the sample average). This increased predicted performance $\hat{y}_i$. The incoming director is currently sitting on two listed boards and 76% of the focal firm's stock is owned by institutions. Both of these attributes increased $\hat{y}_i$.

A *Excess votes as a measure of (better) director performance*



B *(Higher than 10%) dissent as a measure of (worse) director performance*



C *Director turnover as a measure of (worse) director performance*

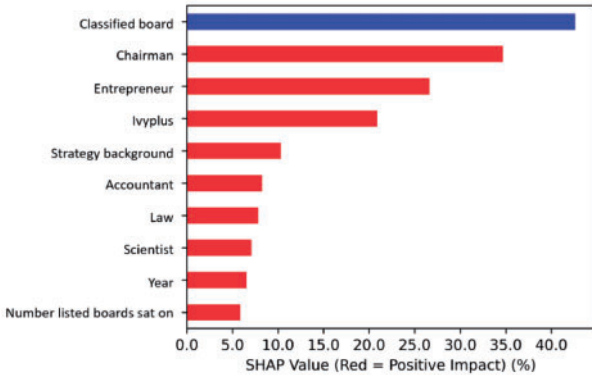


Figure 5
Variable importance plot

This figure presents the SHAP values for the top-10 characteristics in terms of variable importance in predicting director performance. We use the *XGBoost* algorithm in predictions. Variables are ranked in decreasing order of importance. Panel A uses excess votes; panel B uses larger than 10% dissent; and panel C uses director turnover within the next 2 years of appointment as a measure of director performance. While higher excess votes represent better performance, higher dissent or turnover represents worse performance. Features in red (blue) are positively (negatively) correlated with the variable the algorithm predicts.



Figure 6
Individual SHAP values for local interpretability

This figure shows the variables that contributed the most to the *XGBoost*-generated prediction for three random individual observations. Panel A is for an observation when the model predicts excess votes; panel B when it predicts dissent; and panel C when it predicts turnover. The “model output value” is the model’s prediction for this specific observation ($\hat{y}_i$). The “base value” is the mean prediction across all observations. Features in red (blue) increase (decrease) the individual prediction $\hat{y}_i$. The value of the attribute is reported for each observation. The arrow’s length for each attribute corresponds to its SHAP value (i.e., longer arrows represent more important attributes for this observation). The difference between the model output value and the base value is equal to the sum across all attributes of all the SHAP values for a given observation.

Interestingly, some attributes sometimes push up $\hat{y}_i$ and sometimes push it down (e.g., classified board). This pattern speaks to the importance of interactions and nonlinearities in making predictions about director performance and to the fact that there is no one-size-fits-all “good” corporate governance. What may be a “bad” attribute, or a value of an attribute that is “too high” for one firm or one director, could be irrelevant or even have a positive effect for another. This supports our rationale for using machine learning in the context of predicting director quality.

5.2 Overvalued director characteristics

In this section, we consider the characteristics of *predictably bad directors*: directors who the model predicted they would do poorly but were chosen anyway, and subsequently did poorly. These characteristics are likely to be features that tend to be overvalued by firms when they select new directors. To do so, we identify directors who were nominated but were of predictably

Table 6
Overvalued director characteristics

	Focal directors with predicted and observed low shareholder support	Promising candidates for this board position	Difference
	Mean	Mean	<i>p-value</i>
Male	0.88	0.83	0.000
Number of qualifications	2.07	2.23	0.000
Ivy League	0.15	0.15	0.960
MBA	0.52	0.49	0.079
Network size	1714	1261	0.000
Total number of listed boards sat on	6.4	2.6	0.000
Total number of unlisted boards sat on	10.6	2.6	0.000
Total current number of boards sitting on	3.1	1.6	0.000
Number previous jobs same industry	0.07	0.10	0.004
Number previous directorships same industry	0.21	0.10	0.000
Busy	0.56	0.15	0.000
Director age	54.4	56.8	0.000
Background academic	0.043	0.014	0.000
Background finance	0.223	0.178	0.000
International work experience	0.142	0.052	0.000

This table reports the mean of director features for directors in our test set (out of sample predictions) whom our *XGBoost* algorithm predicted would be in the bottom decile of performance and indeed ended up in the bottom decile of actual performance (i.e., predictably low-quality directors) and compares it to the mean for potential candidates the board could have nominated instead, whom our *XGBoost* algorithm predicted would be in the top decile.

low quality, and we compare them to those directors the algorithm would have recommended for that specific board position.²⁴ The patterns of discrepancies between these two groups reflect the types of directors who tend to be overvalued in the nomination process.

Table 6 reports characteristics of these predictably bad directors. Compared to promising candidates identified by the algorithm, predictably unpopular directors are on average more likely to be male, have a larger professional network and more current and past directorships.

These results highlight the features that are likely overrated by the board when nominating directors. They are consistent with the view that men who have sat on a lot of boards tend to be chosen as directors in other firms. The underlying reason for this pattern, however, is not clear. As suggested by the literature on boards going back to Smith (1776) and Berle and Means (1932), managers and existing directors could implicitly collude to nominate new directors unlikely to rock the boat and upset the rents managers and existing directors receive from their current positions. Alternatively, there could be agency problems within boards themselves, that do not involve managers, but in which incumbents want to protect their jobs and knowingly appoint underperforming directors. A third possibility comes from a long literature in psychology dating to Meehl (1954) and highlighted in Kahneman (2011), which has found that even simple algorithms can outperform interviews by trained professionals at predicting

²⁴ Predictions for candidates assume the same committee assignments as the nominated director. We find similar results for all alternative specifications mentioned in the previous sections.

Table 7
Who hires predictably bad directors

	(1)	(2)	(3)	(4)	(5)	(6)
E-index	0.239*** (2.620)	0.245* (1.946)	0.265** (2.070)	0.296** (2.193)	0.452*** (2.695)	0.552*** (2.682)
% co-opted directors		1.238*** (2.767)	1.102** (2.427)	1.107** (2.446)	1.102** (2.122)	1.196** (2.099)
% board independent			−3.627*** (−2.908)	−3.908*** (−3.007)	−3.106* (−1.903)	−4.044** (−1.991)
% board busy				0.616 (0.889)	0.361 (0.397)	0.365 (0.348)
Average board tenure					−0.057 (−1.322)	−0.067 (−1.251)
Number of institutional owners					−0.429 (−1.284)	0.853 (1.216)
Board size					−0.071 (−0.939)	−0.014 (−0.166)
Firm age					−0.005 (−0.421)	−0.005 (−0.311)
ln(assets)						−0.727** (−2.085)
MB						−0.009 (−0.446)
Leverage						1.319 (1.237)
ROA						−0.420 (−0.320)
Excess 12 month-returns						−0.849 (−1.643)
Constant	−3.413*** (−9.863)	−4.188*** (−6.994)	−1.194 (−1.067)	−1.218 (−1.084)	1.056 (0.508)	−0.748 (−0.271)
Observations	3,596	1,775	1,773	1,773	1,713	1,693
Pseudo R-squared	0.0309	0.109	0.174	0.180	0.266	0.353

This table reports results from a probit regression where the dependent variable is a dummy variable equal to one for directors flagged by *XGBoost* as most likely to face strong dissent (decile 10 of dissent predictions) and ended up facing strong dissent on various firm-level governance measures and other board-, and firm-level controls.

subsequent performance in a number of contexts. It is possible that managers and boards could be attempting to find value-maximizing directors but because of behavioral biases, underperform the algorithms we present.

5.3 Why do firms pick predictably bad directors?

The pattern that some firms regularly choose directors who can be predicted to do a poor job suggests that corporate governance is problematic at these firms. Our model provides a way to address this issue directly, by relating a firm’s susceptibility to hiring predictably bad directors with measures of corporate governance. To measure the effect of corporate governance on director choices, Table 7 provides estimates of probit equations that estimate the likelihood that a firm hires a predictably bad director.

As a measure of governance quality, we first use Bebchuk et al.’s (2009) Entrenchment Index (E-index). The index is based on six governance attributes—staggered boards, limits to shareholder by-law amendments, supermajority requirements for mergers, supermajority requirements for charter amendments, poison pills, and golden parachutes—and is constructed so that

it increases as the firm-level governance worsens. This E-index is constructed by evaluating the relative importance of the Investor Responsibility Research Center's (IRRC) 24 provisions, which Gompers, Ishii, and Metrick (2003) include in their G-index. As highlighted in Bebchuk et al. (2009), the first four provisions limit shareholder voting power and thus undermine their ability to impose their will on management. Therefore, we would like to test whether firms whose governance structure disempowers shareholders (as captured by a high E-index) are more likely to select directors who will predictably receive low support from shareholders.

The estimates in Table 7 indicate a statistically significant positive relationship between a firm's E-index and its likelihood of selecting predictably bad directors. We interpret this finding as suggesting that bad choices of directors are related to firms' corporate governance structure: firms more likely to disempower shareholders appear to have objectives other than representing shareholders' interests when selecting directors. Moreover, confirming this interpretation of the results, we find that governance variables are *not* significant when examining the likelihood of selecting a director who received low support, but this was *not* predicted by the algorithm (unpredictably bad directors).

A second measure of governance quality is the percentage of *co-opted* directors, measured as the fraction of the board directors appointed after the CEO assumed office. Coles et al. (2014) provide results suggesting that board monitoring decreases as co-option increases, so co-option likely reflects a board that does not act in the interest of shareholders. We do find a positive and significant relation between predictably bad directors and the percentage of co-opted directors.

Finally, we study whether the fraction of independent directors on a board affects the choice of new directors. With this idea in mind, Congress passed the Sarbanes Oxley Act in 2002, requiring exchange-listed firms to have a majority of independent directors. Consistent with the agency view, the estimates in Table 7 imply that the likelihood of a firm selecting a predictably bad director decreases with the fraction of independent directors on the board.

Overall, the results in Table 7 support the view that agency conflicts distort nominating decisions and clarify why some firms select predictably bad directors.

6. Summary and Discussion of Future Uses of Algorithms in Corporate Governance

We employ machine learning algorithms to predict director performance and show that algorithms could potentially help firms choose directors for their boards. In employing these machine learning algorithms, we contribute to our understanding of corporate governance, specifically boards of directors, in at least four ways. First, we evaluate the possibility of constructing an algorithm that can predict whether a particular individual will be successful as a director

in a particular firm. Second, we compare alternative approaches to forecasting director performance; in particular, how machine learning techniques compare with OLS to predict director performance. Third, we identify characteristics that tend to be associated with effective directors. Finally, we use the selections from the algorithms as benchmarks to understand the process through which directors are actually chosen and identify the types of individuals who are more likely to be poor directors but who are nonetheless selected.

There are a number of methodological issues we must address before we can estimate the quality of a particular director. We must be able to measure the performance of a director to predict which potential directors will be of highest quality. Our main measure of director performance comes from the level of support a director receives from shareholders relative to other directors at the same firm. This vote-based performance measure is an *individual* measure which reflects the support the director personally has from the shareholders she represents and which should incorporate all publicly available information about her performance. We use alternative measures of director performance: the firm's abnormal returns at the time of the announcement of a director's appointment, high dissent, and turnover shortly after the appointment.

Our algorithm does predict the distribution of outcomes using these different measures. However, an issue we need to consider is that we observe the predictive accuracy of our algorithm only for directors who were nominated. To examine this issue, we use the distribution of votes plausible candidates received at the company whose board they joined as an indication of the distribution of votes these candidates would have had on the focal board. Our results suggest that directors the algorithm predicted would do poorly (well) indeed rank poorly (well) not only in absolute terms, but also when compared to realistic alternatives.

The differences between the directors suggested by the algorithm and those actually selected by firms allow us to identify individual features that are overrated in the director nomination process. Comparing predictably bad directors to promising candidates suggested by the algorithm, it appears that predictably bad directors are more likely to be male, have a large network, and have many past and current directorships. A reasonable interpretation of our results is that the algorithm echoes what some institutional shareholders have been saying for a long time: that directors who are not as well known by management and come from different backgrounds are more likely to monitor management. For example, TIAA-CREF (now TIAA) has had a corporate governance policy aimed in large part at diversifying boards of directors since the 1990s for this reason (see Biggs 1996; Carleton, Nelson and Weisbach 1998).²⁵

²⁵ Similarly, Glenn Kelman, the CEO of Redfin, recently wrote: "Redfin has recently completed a search for new board directors, [...] and we had to change our process, soliciting many different sources for candidates rather than relying exclusively on board members' connections. If you don't pay attention to diversity, you'll end up

A natural question concerns the applicability of algorithms, such as the ones we used, in practice and the future uses of algorithms in corporate governance. We view our exercise as a “first-pass” approach, aimed at bringing the predictive power of machine learning tools to the issue of director selection. The performance of machine learning prediction methods has historically improved with the availability of more data and more efficient algorithms. Consequently, we conjecture that more sophisticated models with richer data are likely to be able to predict individual director performance better than the models presented here. If algorithms, such as these, are used in practice in the future, as we suspect they will be, practitioners will undoubtedly have access to much better data than we have used and should be able to predict director performance more accurately than we do in this paper.

An important benefit of algorithms is that they are not prone to the agency conflicts that occur when boards select new directors. Institutional investors are likely to find the algorithm’s independence from agency conflicts appealing and are likely to use their influence to encourage boards to rely on an algorithmic decision aid, such as the one presented here, for director selections in the future.

An important advantage of algorithms over the way in which directors have been chosen historically is that “algorithms can overcome the harmful effects of cognitive biases” (Sunstein, 2018). Nevertheless, algorithmic bias and algorithmic fairness represent concerns of growing importance. Our data, as is almost always the case in the social sciences, were generated by human decisions. As a result, although not inevitable (Rambachan and Roth 2020), machine learning algorithms could possibly generate bias amplification (Barocas and Selbst 2016; Zhao et al. 2017). While our choice of specified outcome helps mitigate the concern of “historical bias” (the decision maker and the evaluator are separate entities), special attention should be paid to various properties of any data set prior to the implementation of an algorithmic decision aid.

Rivera (2012) studies the hiring practices of top investment banks, consulting and law firms and concludes that recruiters overvalue personal fit which is not necessarily a function of expected performance. In the context of lower-skill workers, Hoffman, Kahn, and Li (2017) find that managers who hire against test recommendations recruit worse-than-average hires. Cowgill (2018) shows that the job-screening algorithm at a software company prefers “nontraditional” candidates. Our results suggest that the same idea applies to the nominating of corporate directors. We conjecture that including algorithmic input to limit (but not strip) discretion and reliance on soft information in these decisions could help minimize agency problems, and thus lead to a modified rank ordering

hiring people who are nearest at hand, who have had similar jobs for decades before. This is how society replicates itself from generation to generation, in a process that seems completely innocuous to those who aren’t the ones shut out” (Frey and Kelman 2016).

of candidates. This could in turn lead to better (and potentially more diverse) directors than the current process (see the arguments in Kleinberg et al. 2019).

On the other hand, if the algorithm omits the attributes of potential directors that are valuable to management, such as specialized knowledge of an industry or government connections, then it potentially could lead to suboptimal solutions. Humans and algorithms both have limits and make different kinds of mistakes; they tend to have uncorrelated errors. Striking the right balance in the division of labor between humans and machines to take advantage of their relative strengths is likely going to be a key challenge for future research.²⁶ In addition, if firms progressively start reshuffling their boards based on algorithmic decision aids, behaviors and decisions will start to change. This would likely have general equilibrium effects that would need to be evaluated.

In this paper, we use recent developments in prediction methods to confirm an observation that dates back over 200 years: the board selection process leads to directors who are not necessarily the best choices to serve shareholders' interests. These methods can, however, in addition to confirming this observation, provide us with the tools to change it. By providing a prediction of performance for any potential candidate, an algorithm could expand the set of potential directors and identify individuals with the skills necessary to become successful directors, who would have otherwise been overlooked. We expect that in the not-too-distant future, algorithms will fundamentally change the way corporate governance structures are chosen, and that shareholders will be the beneficiaries.

Appendix A. *Lasso* results

This appendix presents figures analogous to those in the paper, except we use the *Lasso* algorithm here rather than the *XGBoost* algorithm.

²⁶ The grounded discussions in Agrawal, Gans, and Goldfarb (2018) expose the issues around the consequences of AI-based decisions.

Table A.1
List of top-10 *Lasso*-selected features (*Excess votes*)

Characteristics	Coefficient
Total number of unlisted boards sat on	-0.0181
SD of incumbent directors' time on board	0.0061
Compensation committee	-0.0051
Fraction owned by institutions	0.0045
Audit committee	0.0045
Busy	-0.0041
Fraction of incumbent directors with law background	0.0040
SD of incumbent directors' time at the company	0.0026
Generation X (born between 1965 and 1980)	-0.0024
CEO has filed certification documents as required under Section 302 of the Sarbanes-Oxley Act of 2002	0.0024

This table reports the 10 features with the largest absolute coefficients as estimated by *Lasso* in predictions of excess votes. Note, however, that these coefficients cannot be interpreted in the same way as OLS coefficients. See Mullainathan and Spiess (2017) for limitations in interpreting *Lasso* results.

Table A.2
List of top-10 *Lasso*-selected features (*Turnover*)

Characteristics	Coefficient
Fraction of incumbent directors with experience as Partner	0.1116
Succession factor	0.0701
Net debt issues	-0.0687
Other Chair	-0.0675
Newly retained earnings	-0.0655
Compensation chair	-0.0639
Governance chair	-0.0608
Nomination chair	-0.0591
President	-0.0577
Audit chair	-0.0540

This table reports the 10 features with the largest absolute coefficients as estimated by *Lasso* in predictions of turnover within 2 years. Note, however, that these coefficients cannot be interpreted in the same way as OLS coefficients. See Mullainathan and Spiess (2017) for limitations in interpreting *Lasso* results.

Table A.3
List of top-10 *Lasso*-selected features (*Dissent*)

Characteristics	Coefficient
Fraction of incumbent directors with university experience	-0.1450
Fraction owned by the top-five institutional investors	0.1034
Auditor is Moss Adams	-0.0920
CEO has not filed Certification Documents as required under Section 302 of SOX	0.0896
Incumbent directors' average number of directorships in the same industry	0.0846
Auditor is Grant Thornton	0.0777
Incumbent directors' average number of jobs in a quoted company	0.0741
Newly retained earnings	-0.0721
Incumbent directors' average number of jobs in the same industry	0.0714
Chairman	0.0659

This table reports the 10 features with the largest absolute coefficients as estimated by *Lasso* in predictions of dissent (shareholder support below 90%). Note, however, that these coefficients cannot be interpreted in the same way as OLS coefficients. See Mullainathan and Spiess (2017) for limitations in interpreting *Lasso* results.

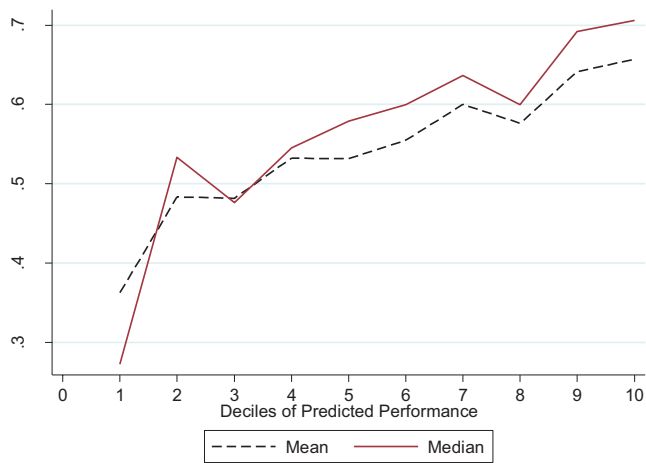


Figure A.1
Mean and median ranks in quasi-label distribution across deciles of *Lasso*-predicted performance
This figure shows the mean and median ranks in the distribution of quasi-labels for directors in each of the 10 deciles of *Lasso*-predicted performance (*Excess votes*). The observed performance of nominated directors in our test set is compared to the quasi-labels of *all* potential candidates in their respective candidate pool: each new board appointment in the test set is associated with a candidate pool, comprising directors who, within 1 year of the appointment, joined the board of a smaller neighboring firm.

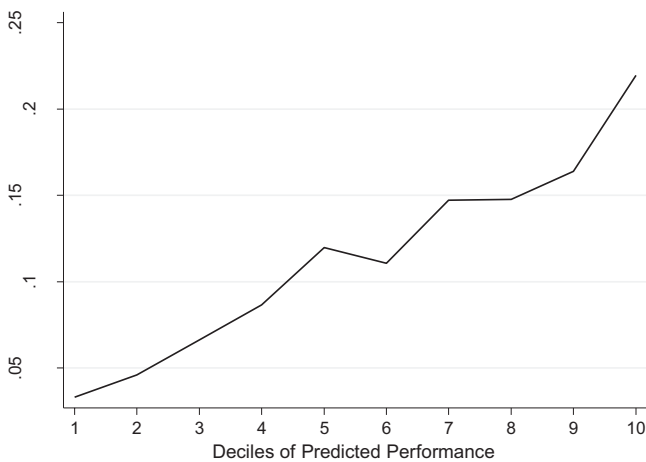


Figure A.2
Mean observed *Dissent* versus *Lasso*-predicted *Dissent*
This figure shows the average observed level of dissent against a director by shareholder (the fraction of new directors who received less than 90% shareholder support in their reelection within the next 3 years) across the 10 deciles of *Lasso*-predicted dissent in the 2012–2014 test set. Directors in decile 1 (10) were predicted by *Lasso* as least (most) likely to face strong dissent. Unconditional mean dissent is about 11%.

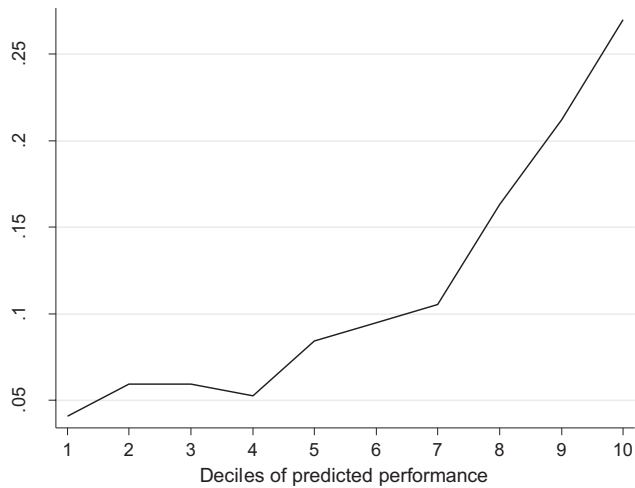


Figure A.3
Mean observed *Director turnover* versus *Lasso-predicted Turnover*
This figure shows the average observed level of director turnover (the fraction of new directors who left within the next 2 years) across the 10 deciles of *Lasso*-predicted turnover in the 2012–2014 test set. The unconditional mean turnover is about 11%.

Appendix B. A Framework to Evaluate Algorithmic Predictions

Appendix B.1. General Framework

We develop a simple framework using the template laid out in Kleinberg et al. (2017) to illustrate how a type of selection bias affects the assessment of the prediction accuracy of algorithms.

Suppose that the true data-generating process is given by $\mathcal{Y} = \mathcal{F}(\mathcal{W}, \mathcal{Z})$, where $\mathcal{W}$ and $\mathcal{Y}$ are operationalized by W , our vector of inputs, and Y , our outcome variable, respectively. $\mathcal{Z}$, operationalized by Z , represents a set of features that affect director performance and that are observable by the decision maker (board/CEO), but not by the algorithm. An example of such a feature would be idiosyncratic knowledge of the firm or its industry that would make a potential director more valuable.

In addition, there are features $\mathcal{B}$, operationalized by B , that do *not* affect director performance and are unobservable to the algorithm, but could nonetheless affect boards' nominating decisions. Examples of such features could be a candidate's political views, or the neighborhood where she grew up. The board's preferences for certain features in $\mathcal{B}$ could be conscious or could represent an implicit bias of which they are unaware of. The key point is that these attributes can influence boards' decisions even though they are *not* correlated with performance.

$\mathcal{F}$ is operationalized by a functional form f . For the purpose of predictive modeling, we are interested in finding a function that closely matches the function f in out-of-sample data. Compared to classic causal hypothesis testing, we do not make strong assumptions about the structure of $\mathcal{F}$ and thus do not focus on examining the estimated parameters and claim that these parameters match f . In other words, our supervised machine learning algorithm seeks to learn a functional form that maps features W into predictions $\hat{f}$ that generalize well on out-of-sample data (Shmueli 2010).

A director is characterized by $\vec{x}$, composed of three vectors of features as well as of outcome y :

$$\vec{x} = \begin{pmatrix} W \\ Z \\ B \end{pmatrix}.$$

Note that, in this case $\vec{x}$ may include not only director characteristics but also firm- and board-level characteristics so that both the board and the algorithm try to assess a director's future performance for a specific board position.

For the purpose of the model and similar to Kleinberg et al. (2017), we shrink the dimension of $\vec{x}$ to a vector with three unidimensional characteristics w , z , and b . In addition, we assume that the sum of w and z is distributed between 0 and 1 and that their sum equals y on average:

$$E[Y=y|W=w, Z=z] = E[y|w, z] = w+z.$$

Each board j has a payoff function π_j that is a function of the director's performance as well as of the director's characteristics as defined by $\vec{x}$.

For each director i in candidate pool $\mathcal{D}$, the board's payoff is characterized as

$$\pi_j(x_i, y_i) = \underbrace{u_j y_i}_{\text{benefits from director's performance}} + \underbrace{v_j g_j(x_i)}_{\text{benefits from hiring director with characteristics } x_i}.$$

$g_j(x)$ is a board specific function that maps directors' characteristics into a score. We can think of $g_j(x)$ as a measure of the utility the board derives from nominating a director with specific characteristics; for example, they could derive private benefits from nominating someone from their own network. The variables u_j and v_j represent the weights that board j puts on director performance and on the benefits the board derives from nominating a director with certain features, respectively.

We assume that board j chooses a nominating rule h_j such that it maximizes its expected payoff.

$$h_j \in \{0, 1\}^{|\mathcal{D}|} \text{ and } \|h_j\|_0 = 1,$$

$$\Pi_j(h_j) = \sum_{i \in \mathcal{D}} h_{j,i} E[\pi_j(x_i, y_i)].$$

We do not observe boards' relative weights on director performance, u_j , and their own preferences for directors with particular characteristics, v_j . In a world of perfect corporate governance, boards are only concerned with their mandate (i.e., representing shareholders' interests) and $v_j = 0$.

We assume $v_j = 0$, not because we believe in a world of perfect governance but because we ask whether an algorithm can identify a director with characteristics x' with better performance than director with characteristics x nominated by board j , whom the board will like at least equally well? In other words, conditional on $g_j(x') \geq g_j(x)$, can an algorithm recommend a nominating rule α that produces a higher payoff than the baseline: the outcome of board j 's actual nominating decision?

The difference in the expected payoffs between the two nominating rules α_j and h_j is

$$\begin{aligned} \Pi_j(\alpha_j) - \Pi_j(h_j) &= \sum_{i \in \mathcal{D}} \alpha_{j,i} E[\pi_j(x_i, y_i)] - \sum_{i \in \mathcal{D}} h_{j,i} E[\pi_j(x_i, y_i)] \\ &= \underbrace{E[y | \alpha_j]}_{\text{missing label}} - \underbrace{E[y | h_j]}_{\text{observed label}}. \end{aligned}$$

We do not observe the performance of directors who would be nominated under the alternative nominating rule produced by the algorithm, $E[y | \alpha_j]$. As discussed in Kleinberg et al. (2017),

missing labels are often dealt with in the machine learning literature by various imputation procedures. However, this approach would assume that if a director shares the same set of observable feature values, w , as the nominated director, their performance would be identical. This is the equivalent of assuming that unobservables, z , play no role in nominating decisions. For a given w , the imputation error would therefore be

$$\begin{aligned} E[y|\alpha_j, w] - E[y|h_j, w] &= E[w+z|\alpha_j, w] - E[w+z|h_j, w] \\ &= E[w|\alpha_j, w] - E[w|h_j, w] + E[z|\alpha_j, w] - E[z|h_j, w] \\ &= E[z|\alpha_j, w] - E[z|h_j, w]. \end{aligned}$$

This imputation error points up the *selective labels problem*. In our setting, it refers to the possibility that directors who were nominated, although they might share the same exact observable features as other directors not nominated, might differ in terms of unobservables. These unobservables could lead to different average outcomes for nominated vs. not nominated, even if both are identical on the basis of observable characteristics. Therefore, it could be the case that even for directors (x, y) who were predictably poor directors, there were no other available director (x', y') such that $\pi_j(x', y') > \pi_j(x, y)$. In other words, there could be supply considerations which make it difficult to evaluate whether algorithmic predictions can help boards improve nominating decisions.

Appendix B.2. A Simple Formalization of the Approach to Evaluate Algorithmic Predictions that Considers Alternative Choices

We examine the challenge described above by constructing a plausible pool $\mathcal{D}_i$ of candidate directors $-i$ for each focal director (x_i, y_i) . We do not observe outcome $E[y|\alpha_j]$ for alternative candidates. We do observe, however, their performance on the board of the (smaller) neighboring firm they joined around the same time, that is, their *quasi-labels* y_{-i} .

Let Y_{-i} be the random variable corresponding to the outcome for the candidates in $\mathcal{D}_i$. We denote $F_{Y_{-i}}(y_i) = P(Y_{-i} \leq y_i)$ director (x_i, y_i) 's percentile rank in the distribution of quasi-labels. While this is a general framework which could accommodate different weighting of some candidates, in practice, we use a uniform distribution over candidates in $\mathcal{D}_i$, $P(Y_{-i} \leq y_i) = \frac{\sum_{-i \in \mathcal{D}_i} \mathbb{I}(y_{-i} \leq y_i)}{|\mathcal{D}_i|}$.

We would like to test whether the algorithm's predictions are informative of future director performance when we consider candidates in the available candidate pool. How did the focal directors perform relative to other available candidates? Is this relative performance correlated with the algorithm's ex ante prediction of performance?

We sort all (x_i, y_i) in our test set into deciles d_n according to their algorithm-predicted performance $\hat{y}$ and estimate the average $P(Y_{-i} \leq y_i)$ for directors in each decile: $\frac{1}{|d_n|} \sum_{(x_i, y_i) \in d_n} P(Y_{-i} \leq y_i) \forall n = (1, \dots, 10)$.

To test whether the algorithm's predictions of director performance are useful even when we consider alternative candidates, we test the null that directors' percentile rank when compared to alternatives is the same for directors predicted to do well as for those predicted to do poorly. This would suggest that although the algorithm achieved high out-of-sample predictive accuracy, unobservables, including the quality of alternative choices, are important. Therefore, the null is that when considering the performance distribution of alternative choices of directors, directors with low predicted performance do as well as those with high predicted performance.

$$H_0: \frac{1}{|d_1|} \sum_{(x_i, y_i) \in d_1} P(Y_{-i} \leq y_i) = \frac{1}{|d_{10}|} \sum_{(x_i, y_i) \in d_{10}} P(Y_{-i} \leq y_i)$$

The alternative is that directors predicted to do poorly rank lower when compared to alternative candidates than those predicted to do well.

$$H_1: \frac{1}{|d_1|} \sum_{(x_i, y_i) \in d_1} P(Y_{-i} \leq y_i) < \frac{1}{|d_{10}|} \sum_{(x_i, y_i) \in d_{10}} P(Y_{-i} \leq y_i)$$

Rejecting the null would suggest that the algorithm's predictions were informative about future performance, even when considering the performance of plausible alternative choices.

Quasi-labels are not perfect substitutes for missing labels. We do not assume that a specific director who was hired by the board of firm B would have necessarily had the same performance on the board of firm A. For each candidate pool associated with a focal director, $\mathcal{D}_i$, we use the distribution of quasi-labels as an ersatz for the distribution of missing labels.

Two assumptions underpin this approach. The first relates to how quasi-labels are collected: they must not depend on algorithmic predictions, $Y_{-i} \perp \hat{y}_i$. How directors in the candidate pools are selected must be independent of the algorithms' predictions of director performance. Directors in the candidate pools are all those hired around the same time and whose board headquarters is nearby, regardless of whether the algorithm predicted the focal director would do well or whether she would do poorly. Some of these hires by nearby firms were likely good choices and some likely were not. But importantly, the appointment process at these firms is similar across prediction deciles; the process used to assign candidate directors in $\mathcal{D}_i$ (and therefore quasi-labels Y_{-i}) does not depend on $\hat{y}_i$.

The second underlying assumption is related to the first and requires that the unobserved difference between the distribution of quasi-labels and the distribution of missing labels is not be systematically negatively correlated with predicted director performance, $(E[y|\alpha_j] - y_{-i}) \perp \hat{y}_i$ ²⁷ We approximate the unobserved distribution of missing labels with the distribution of quasi-labels, and we assume that there is no systematic bias in this approximation (an error systematically correlated with $\hat{y}_i$).

²⁷ Suppose the "error" for the distribution was systematically decreasing as predicted performance increases. For example, suppose the true distribution of missing labels is shifted to the left (right) relative to the distribution of quasi-labels for low (high) levels of predicted performance. In this example, a finding that directors in the bottom (top) decile of predicted performance rank low (high) relative to alternative candidates could potentially be driven, at least in part, by this positive (negative) difference between the true labels and the quasi-labels.

Table C.1
OLS models

Dependent variable	(1) Excess votes	(2) Excess votes	(3) Total votes	(4) Excess votes	(5) Excess votes	(6) Total votes	(7) Excess votes	(8) Excess votes	(9) Total votes	(10) Excess votes	(11) Excess votes	(12) Total votes	(13) Excess votes	(14) Excess votes	(15) Total votes
Compensation chair	-0.007 (-1.307)	-0.007 (-1.264)	0.001 (0.086)	-0.001 (-0.475)	-0.005 (-0.607)	-0.005 (-1.349)	-0.001 (-0.338)	-0.001 (-0.377)	-0.006 (-0.780)	-0.001 (-0.153)	-0.001 (-0.271)	-0.003 (-0.399)	0.001 (0.344)	0.001 (0.279)	-0.002 (-0.535)
Audit chair	-0.006 (-1.089)	-0.006 (-1.245)	-0.005 (-0.480)	0.005*** (2.779)	0.009*** (2.742)	0.009*** (2.731)	0.001 (0.376)	0.001 (0.318)	0.005 (0.681)	0.003 (0.811)	0.003 (0.793)	0.004 (0.571)	0.005*** (2.716)	0.005*** (2.865)	0.009** (2.422)
Gender	-0.002 (-0.843)	-0.002 (-0.847)	-0.000 (-0.007)	-0.001 (-0.631)	-0.001 (-0.562)	-0.001 (-0.216)	0.002 (0.968)	0.002 (1.014)	-0.001 (-0.204)	0.002 (0.978)	0.002 (1.163)	-0.000 (-0.120)	0.000 (0.007)	0.000 (0.019)	-0.004 (-1.525)
Number current other boards	-0.002** (-2.328)	-0.002** (-2.362)	-0.003 (-1.382)				-0.004*** (-6.440)	-0.004*** (-6.763)	-0.003*** (-3.719)	-0.003*** (-4.829)	-0.003*** (-5.315)	-0.003*** (-3.145)			
Director age > 65	-0.003 (-0.778)	-0.002 (-0.525)	0.001 (0.159)				0.001 (0.503)	0.001 (0.608)	-0.001 (-0.284)	-0.000 (-0.031)	0.001 (0.251)	0.001 (0.113)	-0.003** (-2.063)	-0.003** (-2.155)	-0.001 (-0.197)
Fraction owned by director	-9.945 (-0.308)	-15.763 (-0.496)	-5.054 (-0.099)												
Governance chair	0.005 (0.392)	0.005 (0.416)	0.011 (0.424)	0.002 (1.009)	0.002 (0.824)	-0.003 (-0.830)	0.006 (0.618)	0.003 (0.344)	0.003 (0.159)	0.002 (0.242)	-0.001 (-0.115)	-0.008 (-0.410)	0.003 (1.329)	0.003 (1.270)	-0.001 (-0.327)
E-index	0.000 (0.062)	-0.000 (-0.055)	-0.006 (-1.407)				0.001 (1.492)	0.001* (1.672)	0.004 (1.134)	0.000 (0.510)	0.000 (0.793)	0.005* (1.683)			
Governance chair * E-index	-0.003 (-0.635)	-0.003 (-0.750)	-0.002 (-0.208)				-0.002 (-0.696)	-0.001 (-0.378)	-0.004 (-0.790)	-0.001 (-0.420)	-0.000 (-0.017)	-0.001 (-0.084)			
Classified board	0.002 (0.719)	0.000 (0.132)	-0.004 (-0.572)							0.002 (1.119)	0.001 (0.681)	-0.007 (-1.635)	0.001 (1.409)	0.001 (0.819)	-0.003 (-1.065)
In(number institutional owners)	0.003* (1.722)	0.005** (2.533)	0.006 (0.720)							0.002 (0.693)	0.001 (0.543)	0.007 (1.040)			
Industry adj. EBITDA	-0.026** (-2.097)	-0.008 (-0.991)	0.010 (0.221)							-0.016* (-1.831)	-0.006 (-0.970)	0.052* (1.886)	0.002 (0.683)	0.003 (1.068)	-0.017 (-1.290)
Industry adj. 12-months returns	0.002 (0.667)	0.001 (0.496)	-0.002 (-0.269)							0.000 (0.154)	-0.003 (-0.053)	0.001 (0.245)	0.001 (0.837)	0.001 (0.910)	-0.002 (-1.257)
Nomination chair				0.002 (33.00)	100.0 (912.0)	69.0 (02.00)	-0.002 (-0.219)	-0.004 (-0.373)	-0.040* (-1.816)	-0.005 (-0.530)	-0.008 (-0.813)	-0.041** (-2.019)	-0.007 (-1.454)	-0.008 (-1.583)	-0.017* (-1.770)
Industry experience															
Background finance															
Background law															
MBA				0.001 (0.810)	0.001 (0.976)	-0.000 (-0.067)	0.000 (0.037)	-0.000 (-0.005)	0.001 (0.343)						
Ivy plus				0.000 (0.040)	0.000 (0.574)	0.001 (0.056)	-0.000 (-0.003)	-0.000 (-0.003)	-0.000 (-0.724)						
Director age				-0.000 (-0.000)	-0.000 (-0.499)	-0.000 (-0.854)	-0.004 (-0.044)	-0.005 (-0.095)							

Table C.1
Continued

Dependent variable	(1) Excess votes	(2) Excess votes	(3) Total votes	(4) Excess votes	(5) Excess votes	(6) Total votes	(7) Excess votes	(8) Excess votes	(9) Total votes	(10) Excess votes	(11) Excess votes	(12) Total votes	(13) Excess votes	(14) Excess votes	(15) Total votes
Number of qualifications															
In(assets)				0.000 (0.324)	0.000 (0.041)	0.002*** (2.816)	-0.000 (-0.440)	-0.005 (-0.605)	0.001 (0.852)	0.001 (0.899)	0.001 (0.836)	0.002 (1.192)	0.000 (0.249)	0.000 (0.219)	0.001* (1.837)
Leverage				0.001*** (2.545)	0.001*** (2.649)	-0.011** (-2.476)	0.001 (0.866)	0.001 (1.195)	-0.004 (-0.328)	0.001 (0.986)	0.001 (1.146)	0.001 (0.077)	0.001*** (2.839)	0.001*** (2.660)	0.001*** (-2.479)
MB				-0.005** (-2.264)	-0.004** (-2.024)	0.025** (2.024)	-0.001 (-0.172)	-0.002 (-0.294)	0.020 (0.633)	0.001 (0.221)	-0.003 (-0.544)	0.047 (1.511)	-0.006** (-2.342)	-0.005** (-2.351)	0.003 (0.214)
Largest 5 institutional owners %				0.000 (-0.132)	0.000 (-0.336)	0.000 (1.062)	-0.000 (-0.303)	-0.000 (-0.384)	-0.000 (-0.317)	-0.000 (-0.139)	-0.000 (-0.208)	-0.000 (0.031)	-0.000 (-0.361)	-0.000 (-0.500)	0.000 (0.536)
ROA				0.005 (1.383)	0.009** (2.490)	-0.049*** (-2.819)	0.017* (1.747)	0.022** (2.449)	-0.075** (-2.201)	0.011 (0.787)	0.016 (1.167)	-0.110** (-2.378)	0.006 (1.497)	0.011*** (2.791)	-0.029 (-1.513)
Product market fluidity				0.002 (0.614)	0.000 (-2.030)	0.000 (0.563)	0.000 (-0.002)	-0.000 (-0.000)	0.005 (1.442)	-0.000 (-0.099)	-0.000 (-0.135)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
12-months returns				0.000 (0.040)	0.000 (0.021)	-0.002 (-1.381)	0.000 (-0.001)	-0.000 (-0.001)	0.005 (1.194)	-0.000 (-0.227)	-0.000 (-0.533)	-0.001 (-0.417)	-0.000 (-0.138)	-0.000 (-0.866)	0.000 (0.129)
Dividend payer				0.001 (0.848)	0.001 (0.504)	0.005 (0.828)	0.000 (-0.375)	0.001 (0.523)	0.014 (1.194)	-0.001 (-0.000)	-0.001 (-0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Board size				-0.000 (-1.081)	-0.000 (-1.560)	-0.000 (-0.489)	-0.000 (-0.130)	-0.000 (-0.391)	-0.000 (-0.002)	-0.000 (-0.286)	-0.000 (-0.227)	-0.001 (-0.417)	-0.000 (-0.138)	-0.000 (-0.866)	0.000 (0.129)
Fraction female on board				0.000 (-1.135)	0.000 (0.315)	0.003 (1.375)	0.000 (0.300)	0.000 (0.228)	-0.002 (-0.652)	0.000 (0.821)	0.000 (0.211)	-0.004 (-0.458)	0.001 (0.864)	0.001 (0.575)	-0.004 (-0.690)
Fraction independent directors				-0.006* (-1.701)	-0.006 (-1.512)	0.010 (0.549)	-0.008 (-0.955)	-0.007 (-0.853)	0.055 (1.393)	-0.013 (-1.501)	-0.012 (-1.415)	0.029 (0.695)	-0.005 (-1.175)	-0.004 (-1.029)	0.013 (0.608)
Average director age				-0.000 (-0.204)	0.000 (0.759)	0.001 (1.110)	-0.000 (-0.331)	0.000 (0.340)	-0.002 (-1.625)	-0.000 (0.011)	-0.000 (0.821)	-0.004 (-0.458)	0.001 (0.864)	0.001 (0.575)	-0.004 (-0.690)
Chairman CEO duality				-0.000 (-0.204)	0.000 (0.759)	0.001 (1.110)	-0.000 (-0.331)	0.000 (0.340)	-0.002 (-1.625)	-0.000 (0.011)	-0.000 (0.821)	-0.004 (-0.458)	0.001 (0.864)	0.001 (0.575)	-0.004 (-0.690)
Fraction co-opted directors				-0.000 (-0.204)	0.000 (0.759)	0.001 (1.110)	-0.000 (-0.331)	0.000 (0.340)	-0.002 (-1.625)	-0.000 (0.011)	-0.000 (0.821)	-0.004 (-0.458)	0.001 (0.864)	0.001 (0.575)	-0.004 (-0.690)
Busy				0.039** (2.184)	0.011*** (2.846)	0.975*** (59.837)	0.010 (0.389)	0.005 (0.329)	1.068*** (8.806)	0.034* (1.748)	0.013 (1.603)	0.935*** (8.982)	-0.006*** (-5.552)	-0.006*** (-5.573)	-0.008*** (-3.656)
Constant				0.039** (2.184)	0.011*** (2.846)	0.975*** (59.837)	0.010 (0.389)	0.005 (0.329)	1.068*** (8.806)	0.034* (1.748)	0.013 (1.603)	0.935*** (8.982)	-0.006*** (-5.552)	-0.006*** (-5.573)	-0.008*** (-3.656)
Observations				1,345	1,345	1,255	10,601	3,136	3,064	3,040	3,040	2,823	8,773	8,773	24,446
R-squared				0.051	0.015	0.632	0.012	0.022	0.668	0.034	0.015	0.662	0.015	0.007	0.602
Calendar year FE				yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Industry FE				yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Firm-Year FE				yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
AIC				-4940	-4993	-4695	-36989	-10956	-11015	-10778	-10826	-10344	-31045	-31092	-28344

This table reports coefficients from various OLS models of excess votes and total votes on various director, firm, and board characteristics. Excess vote is defined as the average observed level of shareholder support over the first 3 years of a new director's tenure, minus the average vote for all directors in the same slate. The regression sample contains director appointments between 2000 and 2011. The Akaike information criterion (AIC), all reported in the last row, is the relative quality of the STS models presented.

References

- Abis, S. 2017. Man vs. machine: Quantitative and discretionary equity management. Working Paper, Columbia University.
- Adams, R. B. 2017. Boards and the directors who sit on them. In *Handbook of the economics of corporate governance*, chap. 6, eds. B. E. Hermalin and M. S. Weisbach. Amsterdam, the Netherlands: Elsevier.
- Adams, R., B. Hermalin, and M. Weisbach. 2010. The role of boards of directors in corporate governance: A conceptual framework and survey. *Journal of Economic Literature* 48:58–107.
- Aggarwal, R., S. Dahiya, and N. Prabhala. 2017. The power of shareholder votes: Evidence from uncontested director elections. Research Paper, Georgetown McDonough School of Business.
- Aggarwal, R., I. Erel, and L. Starks. 2016. Influence of public opinion on investor voting and proxy advisors. Working Paper, Ohio State University.
- Agrawal, A., J. Gans, and A. Goldfarb. 2018. *Prediction machines*. Boston, Massachusetts: Harvard Business Review Press.
- Athey, S. 2017. Beyond predictions: Using big data for policy problems. *Science* 355:483–5.
- Athey, S., and G. W. Imbens. 2017. The state of applied econometrics: Causality and policy evaluation. *Journal of Economic Perspectives* 31:3–32.
- Barocas, S., and A. Selbst. 2016. Big data's disparate impact. *California Law Review* 104:671–732.
- Bates, T., D. Becher, and J. Wilson. 2017. Performance-Based Turnover on Corporate Boards. Working paper.
- Bebchuk, L. A. Cohen, and A. Ferrel. 2009. What matters in corporate governance? *Review of Financial Studies* 22:783–827.
- Berle A., and G. Means. 1932. *The modern corporation and private property*. New York: Commerce Clearing House.
- Biggs, J. 1996. Corporate governance assessment: A TIAA-CREF initiative. *Directors Monthly* 20:1–8.
- Boone, A., L. Field, and J. Karpoff. 2007. The determinants of corporate board size and composition: an empirical analysis. *Journal of Financial Economics* 85:66–101.
- Bubb, R., and E. Catan. 2020. The party structure of mutual funds. Working Paper, ECGI Law.
- Cai, J., J. Garner, and R. Walkling. 2009. Electing directors. *Journal of Finance* 64:2389–421.
- Cai, J., T. Nguyen, and R. Walkling. 2017. Director appointments – it is who you know. Working Paper, Drexel University.
- Carleton, W. T., J. M. Nelson and M. S. Weisbach. 1998. The influence of institutions on corporate governance through private negotiations: Evidence from TIAA-CREF. *Journal of Finance* 53:1335–62.
- Chen, T., and C. Guestrin. 2016. XGBoost: A scalable tree boosting system. In *KDD '16 Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–94. New York: ACM.
- Coles, J., N. Daniel, and L. Naveen. 2008. Boards: Does one size fit all? *Journal of Financial Economics* 87:329–56.
- . 2014. Co-opted boards. *Review of Financial Studies* 27:1751–96.
- Cowgill, B. 2018. Automating judgement and decision-making: Theory and evidence from resumé screening. Working Paper, Columbia Business School.
- Easley, D., M. Lopez de Prado, M. O'Hara, and Z. Zhang. 2020. Microstructure in the machine age. *Review of Financial Studies* 34:3316–63.

- Edmans, A. 2012. The link between job satisfaction and firm value, with implications for corporate social responsibility. *Academy of Management Perspectives* 26:1–19.
- Erel, I., L. Stern, C. Tan, and M. S. Weisbach. 2018. Selecting directors using machine learning. Working Paper, Ohio State University.
- Ertimur, Y. F. Ferri, and D. Oesch. 2018. Shareholder votes and proxy advisors: Evidence from say on pay. *Journal of Accounting Research* 51:951–96.
- . 2018. Understanding uncontested director elections. *Management Science* 64:3400–3420.
- Fedaseyeu, V., J. Linck, and H. Wagner. 2018. Do qualifications matter? New evidence on board functions and director compensation. *Journal of Corporate Finance* 48:816–39.
- Ferreira, D., E. Ginglinger, M. Laguna, and J. Skalli. 2017. Closing the gap: Gender quotas and corporate board composition. Working Paper, ECGI
- Fich, E., and A. Shivdasani. 2006. Are busy boards effective monitors? *Journal of Finance* 61:689–724.
- Field, L., M. Souther, and A. Yore. 2020. At the table but can't break through the glass ceiling: Board leadership positions elude diverse directors. *Journal of Financial Economics* 137:787–814.
- Fischer, P., J. Gramlich, B. Miller and H. White. 2009. Investor perceptions of board performance: Evidence from uncontested director elections. *Journal of Accounting and Economics* 48:172–89.
- Fos, V., K. Li, and M. Tsoutsoura. 2018. Do director elections matter? *Review of Financial Studies* 31:1499–531.
- Frey, B., and G. Kelman. 2016. How to triple the number of women appointed to boards in three years. *Redfin News*, November 16. <https://www.redfin.com/news/how-to-triple-the-number-of-women-appointed-to-boards-in-three-years/>
- Gompers, P., J. Ishii, and A. Metrick. 2003. Corporate governance and equity prices. *Quarterly Journal of Economics* 118:107–55.
- Gormley, T., and D. Matsa. 2014. Common errors: How to (and not to) control for unobserved heterogeneity. *Review of Financial Studies* 27:617–61.
- Hart O., and L. Zingales. 2017. Companies should maximize shareholder welfare not market value. *Journal of Law, Finance, and Accounting* 2:247–74.
- Hastie, T., R. Tibshirani, and J. Friedman. 2009. *The elements of statistical learning*. New York: Springer.
- Heath, D., D. Macciocchi, R. Michaely, and M. Ringgenberg. 2019. Passive investors are passive monitors. Working Paper, University of Utah.
- Hermalin, B. E., and M. S. Weisbach. 1998. Endogenously chosen boards of directors and their monitoring of the CEO. *American Economic Review* 88:96–118.
- . 2003. Boards of directors as an endogenously determined institution: A survey of the economic literature. *Federal Reserve Bank of New York Economic Policy Review* 9:7–26.
- Hoberg, G., and G. Phillips. 2010. Product market synergies and competition in mergers and acquisitions: A text-based analysis. *Review of Financial Studies* 23:3773–811.
- . 2016. Text-based network industries and endogenous product differentiation. *Journal of Political Economy* 124:1423–65.
- Hoffman, M., L. B. Kahn, and D. Li. 2018. Discretion in hiring. *Quarterly Journal of Economics* 133:765–800.
- Iliev, P., K. Lins, D. Miller and L. Roth. 2015. Shareholder voting and corporate governance around the world. *Review of Financial Studies* 28:2167–202.
- Iliev, P., and M. Lowry. 2014. Are mutual funds active voters? *Review of Financial Studies* 28:446–85.
- Kahneman, D. 2011. *Thinking, fast and slow*. New York: Farrar, Strauss and Giroux.

- Ke, Z., B. Kelly, and D. Xiu. 2019. Predicting returns with text data. Working Paper, Harvard University.
- Kleinberg, J., H. Lakkaraju, J. Leskovec, J. Ludwig, S. Mullainathan. 2017. Human decisions and machine predictions. *Quarterly Journal of Economics* 133:237–93.
- Kleinberg, J., J. Ludwig, S. Mullainathan, and C. Sunstein. 2019. Discrimination in the age of algorithms. *Journal of Legal Analysis* 10:113–74.
- Kramarz, F., and D. Thesmar. 2013. Social networks in the boardroom. *Journal of the European Economic Association* 11:780–807.
- Li, K., M. Feng, R. Shen, and X. Yan. 2020. Measuring corporate culture using machine learning. *Review of Financial Studies* 34:3265–315.
- Linck, J., J. Netter, and T. Yang. 2008. Determinants of board structure. *Journal of Financial Economics* 87:308–28.
- . 2009. The effects and unintended consequences of the Sarbanes-Oxley Act on the supply and demand for directors. *Review of Financial Studies* 8:3287–328.
- Lundberg, S., and S. Lee. 2017. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, 4768–77. New York: ACM.
- Masulis, R., and S. Mobbs. 2014. Independent director incentives: Where do talented directors spend their limited time and energy. *Journal of Financial Economics* 111:406–29.
- Meehl, P. E. 1954. *Clinical vs. statistical prediction: A theoretical analysis and a review of the evidence*. Minneapolis: University of Minnesota Press.
- Mullainathan, S., and J. Spiess. 2017. Machine learning: An applied econometric approach. *Journal of Economic Perspectives* 31:87–106.
- Peters, R. H., and L. A. Taylor. 2017. Intangible capital and the investment-q relation. *Journal of Financial Economics* 123:251–72.
- Rambachan, A., and J. Roth. 2020. Bias in, bias out? Evaluating the folk wisdom. arXiv, preprint, <https://arxiv.org/abs/1909.08518>.
- Ribeiro, M., S. Singh, and C. Guestrin. 2016. “Why should i trust you?” Explaining the predictions of any classifier. arXiv, preprint, <https://arxiv.org/abs/1602.04938>.
- Rivera, L. 2012. Hiring as cultural matching: The case of elite professional service firms. *American Sociological Review* 77:999–1022.
- Rossi, A. 2018. Predicting stock market returns with machine learning. Working Paper, University of Maryland.
- Shapley, L. S. 1951. *Notes on the n-person game – II: The value of an n-person game*. Santa Monica, CA: RAND.
- Shivdasani, A., and D. Yermack. 1999. CEO Involvement in the selection of new board members: An empirical analysis. *Journal of Finance* 54:1829–53.
- Shmueli, G. 2010. To explain or to predict? *Statistical Science* 25:289–310.
- Smith, A. 1776. *An inquiry into the nature and causes of the wealth of nations*. Indianapolis, IN: Liberty Press.
- Sunstein, C. 2019. Algorithms, correcting biases. *Social Research* 86:499–51.
- Vilone, G., and L. Lungo. 2020. Explainable artificial intelligence: A systematic review. arXiv, preprint, <https://arxiv.org/abs/2006.00093>.
- Zhao, J., T. Wang, M. Yatskar, V. Ordonez, and K.-W. Chang. 2017. Men also like shopping: Reducing gender bias amplification using corpus-level constraints. arXiv, preprint, <https://arxiv.org/abs/1707.09457> EMNLP.